

Information Gain Prefers the Questions Users Cannot Answer: Reliability-Aware Query Selection for Interactive Identification

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Abstract—Interactive identification systems—troubleshooters, triage assistants, product finders, twenty-questions engines—choose what to ask next by maximising expected information gain, a criterion derived under the assumption that answers are observed exactly. Human answers are not. We show that the two properties governing a question’s value, how evenly it splits the belief and how reliably a person can answer it, are not independent, and that in a 79,803-track music catalogue their dependence runs in the unfavourable direction: a classical selector asks questions whose annotated error rate averages 0.225 against a catalogue average of 0.098, spending 54% of a twenty-question budget on the attributes users judge worst, while a random selector faces 0.097, the catalogue average. We trace the effect to preprocessing rather than to the domain. Quantile binning fixes an attribute’s maximum one-versus-rest entropy at $H_b(1/k)$ regardless of its semantics, so discretised perceptual descriptors are assigned peak apparent informativeness; refining the binning to four equal-population bins strengthens the measured bias from $\rho = -0.44$ to $\rho = -0.61$, and the effect is absent, as this account predicts, in two catalogues that contain nothing to discretise. The correction is one line: replace the entropy of the induced split with the mutual information between target and *observed* answer under a per-attribute binary symmetric channel. We claim no novelty for that quantity—it is textbook—but we show it re-ranks rather than rescales, that a uniform noise assumption leaves the selector provably unchanged, and that a learned value-regression policy with sufficient feature capacity fails to recover the weighting and gets worse, not better, when noise is added to its training rollouts. On 900 paired trials with exact McNemar testing, at a forty-question budget on the full catalogue, reliability-aware selection raises top-1 accuracy from 16.56% to 26.56% and top-5 from 27.22% to 44.78% ($p = 2.0 \times 10^{-9}$ after Holm correction), moving the median rank of the true item among 79,803 candidates from 59.0 to 8.8. The advantage widens rather than narrows with budget, and survives every operating point we tested: over a grid of six catalogue sizes and three budgets it is positive in all eighteen cells, by 6.5 to 32.2 percentage points, and at a thousand-item catalogue with forty questions reliability-aware selection identifies the target 84.2% of the time. Crucially, the result does not depend on the three assumed reliability values. Over 60 worlds drawn from priors on them, a system that ships the fixed point estimates wins in 60 of 60, with a minimum advantage of 1.3 percentage points; the annotation may be wrong for a majority of attributes before the correction stops paying; and the values need not be assumed at all. Treating the target item as a latent variable and running EM over 600 logged sessions of the incumbent system recovers a per-attribute crossover vector with no annotation, no labels and no user study, and a system shipping that learned vector matches the hand-annotated one exactly (8.33% against 8.33%,

38 discordant pairs each way). We also report where the method fails—clean channels, scrambled or inverted annotations—and give a two-step test, computable from a catalogue with no users at all, that predicts in advance whether reliability-aware selection will pay.

Index Terms—Interactive identification, active learning, information gain, query selection, noisy oracles, conversational recommendation, binary symmetric channel, discretisation.

I. INTRODUCTION

A RECURRING interaction pattern places a machine in the role of questioner and a person in the role of oracle. A troubleshooting assistant narrows a fault by asking about symptoms; a clinical triage tool narrows a differential by asking about presentation; a product finder narrows a catalogue by asking about requirements; the parlour game Akinator narrows a space of characters by asking about traits. In each case the system holds a belief over a discrete set of hypotheses, selects a question, observes an answer, and revises. The engineering question is which question to ask, and the near-universal answer is the one that maximises expected information gain.

That criterion inherits its authority from a clean theoretical setting in which the answer is observed exactly. Human answers are not observed exactly. A user asked whether the song they have in mind is “energetic” may hesitate, may apply a personal threshold, or may simply be wrong, and the system has no way to distinguish a considered answer from a guess. What makes this more than a routine robustness concern is that the unreliability is not spread uniformly over the question space. Some attributes admit near-certain answers: a listener generally knows the performing artist and the language of the lyrics. Others are irreducibly fuzzy, and inter-annotator agreement on affective music descriptors is documented to be modest even among trained annotators [1], [2].

A. The Problem

The observation that motivates this work is that the two properties—how well a question divides the belief, and how reliably it can be answered—need not be independent, and that in at least one large real catalogue their dependence runs in the unfavourable direction. An attribute is vague when competent people disagree about where its boundary falls. Disagreement distributes items on both sides of that boundary. A near-even split is exactly what information gain rewards. The criterion

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Reference implementation, catalogues and the complete result files needed to reproduce every number in this paper are available at [repository URL].

can therefore acquire a structural tendency to promote the questions that are hardest to answer, and it does so silently, because nothing in the objective refers to answerability.

Routine preprocessing sharpens the effect. Continuous descriptors are commonly discretised at quantiles so that they can be posed as binary questions. A median split is a 50/50 split by construction, which is the maximum of the binary entropy function. The discretisation step thus assigns peak apparent informativeness to continuous perceptual features, which are the features users answer worst, while high-cardinality or heavily skewed identifiers generate only lopsided, low-entropy binary questions and are deprioritised despite being answerable with near certainty.

B. What We Claim, and What We Do Not

We are careful about the scope of the empirical claim, because our own measurements narrow it. On a 79,803-track music catalogue the bias is present and statistically detectable: attributes annotated as reliably answerable carry roughly half the maximum information gain of the rest. On two further catalogues drawn from different domains—mushroom identification and dermatological triage—it is absent. That is not a negative result to be buried; it is a confirmation of the proposed mechanism, because neither of those catalogues contains a continuous descriptor to discretise, and the mechanism we blame is discretisation acting on perceptual quantities. The honest generalisation is therefore conditional, and we state it as a scope condition rather than as a universal: the bias should be expected wherever a catalogue mixes quantile-binned perceptual descriptors with naturally skewed objective metadata, and should not be expected where every attribute is natively categorical.

The absence of the bias turns out not to settle whether the correction is worth making, which is the most useful thing the cross-domain experiment taught us. On dermatology, where the bias is absent, reliability-aware selection more than doubles top-1 accuracy; on mushroom, where it is equally absent, the correction changes nothing at all. What separates them is not the correlation but whether the answerable attributes still carry enough information to identify the target—and that is a property of the catalogue, computable before any user is involved.

We are equally careful about the acquisition function. Correcting the criterion means replacing the entropy of the induced split with the mutual information between the target and the *observed* answer, under an explicit noise model. For a per-attribute binary symmetric channel that quantity is $H_b(p_q \star \varepsilon_q) - H_b(\varepsilon_q)$, which is textbook [3, Ch. 7] and which we claim no novelty for. What we claim is empirical and diagnostic: that the observation model in some attribute catalogues is heterogeneous in a specific and unfavourable way, that this is partly an artefact of preprocessing rather than of the domain, that a learned selection policy trained on simulated interaction inherits the same blind spot, and that correcting it requires far less knowledge about users than one might expect—in the end, none that a deployed system cannot recover from its own interaction logs.

C. On the Operating Point

Two parameters govern how hard this task is, and neither is a property of any method: how many questions the system may ask, and how many items it must choose between. Both need stating in advance, because absolute accuracies mean nothing without them and because a reader is entitled to know they were not chosen to flatter.

Budget. The catalogue has an entropy floor of 16.23 bits, and the capacity-corrected bound of Corollary 1 puts the minimum sufficient budget at 20.15 questions under the reliability annotation used throughout. A system run at $T = 20$ is therefore operating essentially *on* the information-theoretic boundary, where even perfect knowledge of the noise barely suffices. That is a stress test, and we use it as one: the condition sweep, and every robustness experiment in Section VIII, is run there. The headline comparison is run at $T = 40$, roughly twice the bound and within the range a deployed identifier would actually spend, where reliability-aware selection reaches 26.6% top-1 and 44.8% top-5 against classical information gain’s 16.6% and 27.2%. We note explicitly that this is the conservative direction to have moved in: the advantage *widens* with budget, so reporting at the floor understated our own result rather than exaggerating it.

Catalogue size. Eighty thousand items is the largest catalogue we have and the hardest version of the task; it is not the version most deployed identifiers face. Because size is a task parameter, we sweep it, and report accuracy over a grid of sizes and budgets (Section VII-F). At a thousand items and forty questions the same method with the same annotation identifies the target 84.2% of the time, and 88.3% if allowed sixty questions. That sweep is emphatically not cross-validation and we do not present it as such—no model is fitted, so a subset does not estimate performance on the whole, it simply poses an easier question, and a number obtained on a subset must be reported with the subset size attached. What it establishes is that the advantage we report is not an artefact of operating near the floor: it is positive in every one of the eighteen cells, by between 6.5 and 32.2 percentage points, and every cell is significant at $\alpha = 0.01$.

Alongside both we report rank-based metrics, which are not compressed against the floor, and the budget each method needs to reach a fixed accuracy, which is the quantity a practitioner actually allocates.

D. Limitations of Existing Approaches

Three responses to answer noise appear in the applied literature, and none addresses the bias described above.

The first is to replace hard candidate elimination with a probabilistic belief update, so that a contradicted candidate is downweighted rather than discarded. This is correct and necessary, and it is where the robustness reported by comparative studies of interactive identification actually originates. It does not change which questions are asked.

The second is to learn a question-selection policy from simulated interaction, typically by regressing a realised measure of progress onto features of the state and the query. This substitutes an approximation for a quantity that has a closed

form under the model, and in the absence of an explicit noise term the learned policy inherits the bias from its training signal. We do not leave this as an assertion: we implement the policy, give it feature capacity sufficient to represent the reliability weighting, train it both with and without noise in its rollouts, and report what it does.

The third is to discard unreliable attributes altogether. This is a reasonable practitioner heuristic and we treat it as a baseline, but it is wasteful: an attribute answered correctly seventy per cent of the time still transmits information, and discarding it removes that information permanently rather than deferring its use to the point where more reliable questions have been exhausted.

E. Research Gap

The theory needed to correct the bias is not new. Noisy search has been studied since Rényi [4] and Ulam [5], surveyed by Pelc [6], and placed on a decision-theoretic footing by Jedynek et al. [7]. Bayesian experimental design supplies the general principle [8], [9], and the active-learning literature has established both that naive greedy selection degrades badly under noisy observations and that noise-aware greedy alternatives with approximation guarantees exist [10], [11]. What is missing is not an algorithm. It is the empirical observation that in some attribute-indexed catalogues the observation model is strongly heterogeneous *across attributes* and adversarially aligned with the naive acquisition function, an account of where that heterogeneity comes from, and a deployable answer to the question of what a practitioner must actually know in order to correct it. The noisy-search literature typically assumes a single known crossover probability; the applied literature typically assumes none at all.

F. Contributions

- 1) **A characterisation of the bias together with its scope condition.** We measure the relationship between per-attribute information gain and per-attribute answerability, using a test appropriate to an annotation with three levels rather than the rank correlation that a three-valued variable inflates the ties of, and we isolate the contribution of the discretisation step. We then test the mechanism by predicting and confirming the absence of the bias in two natively categorical catalogues (Sections VII-A and VIII-F).
- 2) **A unification that removes a standing confound.** Hard elimination is the $\varepsilon \rightarrow 0$ limit of the Bayesian update (Proposition 1), so comparative studies that pair an information-gain selector with elimination and a learned selector with soft updating vary two factors at once. We separate them explicitly (Section VI-C).
- 3) **Reliability-aware information gain.** We derive $\text{RAIG}(q) = H_b(p_q \star \varepsilon_q) - H_b(\varepsilon_q)$, show that it re-ranks rather than rescales (Proposition 3), derive a capacity-based lower bound on the query budget (Corollary 1), formalise the discretisation mechanism (Proposition 4), and extend the criterion to admit

abstention (Section III-G). The change is one line in an existing selector.

- 4) **A crossover vector learned from interaction logs, and a robustness result that needs none of the assumed values.** The obvious objection to this work is that its three reliability constants are invented. We answer it three ways. We re-run the primary comparison over an ensemble of worlds drawn from priors on those constants, so the conclusion is reported across plausible settings rather than at one point; we corrupt the annotation and measure how much of it may be wrong before the correction stops paying; and—the substantive answer—we show the constants are unnecessary. Treating the target item as latent and running EM over the logs of an already-deployed incumbent recovers a per-attribute crossover vector with no annotation, no labels and no user study, and a system shipping that vector matches the hand-annotated one exactly (Sections IV-C, VIII-A, VIII-B and VIII-E).
- 5) **A learned-policy comparison.** We implement, train and evaluate the learned-selector family the literature would otherwise supply only as a citation, and report the extent to which it does and does not recover reliability-aware behaviour (Section VII-D).
- 6) **A deployment test that requires no users.** The cross-domain results separate two properties that are easily conflated. Whether informativeness correlates with unanswerability decides whether classical selection is being actively misled; whether the answerable attributes still carry enough information to finish the job decides whether a reliability-aware selector can do anything about it. The second is computable from a catalogue in seconds, by counting distinct attribute vectors over the reliable subset, and it—not the correlation—predicts which of our three catalogues benefits (Section VIII-G).
- 7) **An open, bitwise-reproducible harness** with paired sampling, exact McNemar testing with multiplicity control, three catalogues, and the restricted-attribute baseline that most directly threatens the method.

II. BACKGROUND AND RELATED WORK

A. Searching with Errors

Identifying an unknown element through queries whose answers may be wrong originates with Rényi’s guessing game in which the responder lies with fixed probability [4], and independently with Ulam’s question of how many yes/no queries suffice to locate an integer below one million when a bounded number of answers may be false [5]. The combinatorial branch, in which lies are adversarial and bounded in number, produces optimal strategies with a direct correspondence to error-correcting codes; Pelc’s survey remains the standard reference [6]. The probabilistic branch treats errors as channel noise and connects to sequential transmission with feedback [12], [13], to noisy binary search [14], [15], and to treatments in which the noise depends on the query itself [16], [17].

Our setting sits in the probabilistic branch, with one departure that turns out to matter. Existing formulations assume a

single crossover probability, or make it depend on a continuous query parameter. In an attribute catalogue the crossover is a property of the semantic category being probed and can vary by an order of magnitude across categories. That heterogeneity, rather than the presence of noise as such, is what drives the effect we report.

B. Active Learning and Experimental Design under Noise

Choosing the experiment that maximises expected information about a latent quantity is the central prescription of Bayesian experimental design [8], [9], and in the noiseless binary case it reduces to maximising the entropy of the induced partition. Geman and Jedynek’s active testing model [18] is an early application of the same principle to sequential narrowing. Jedynek et al. [7] give the result most often invoked here: for twenty questions with noise under entropy loss, a greedy one-step policy is optimal over the whole horizon.

That result is frequently cited as licence to select greedily and stop worrying, and it is worth being precise about what it does and does not cover. It is established under a stated observation model with a single noise level; it is not a statement about the heterogeneous, per-attribute channel studied here. The active-learning literature is in fact explicit that noise breaks naive greedy selection: Golovin, Krause and Ray show that generalised binary search, near-optimal without noise, “can perform very poorly” once observations are noisy, and give EC², a greedy algorithm over equivalence classes with a competitiveness guarantee under noise [10]; the adaptive-submodularity framework supplies the general machinery [11]. Our position relative to this work should be stated plainly, since it is the closest technical prior art. EC² and its relatives take the noise model as given and construct a selection rule with guarantees against it. We take the selection rule as given — greedy mutual information, which is what deployed systems actually use — and make an empirical claim about the *shape* of the noise model in real attribute catalogues, namely that it is heterogeneous across attributes, partly manufactured by preprocessing, and aligned against the naive criterion. The two are complementary: a practitioner adopting EC² would still need to know the per-attribute crossover probabilities, and Section IV is about how to get them.

The broader active-learning literature [19] treats oracle fallibility under the heading of proactive learning, where oracles differ in cost and reliability and the learner chooses among them [20], and under label noise more generally [21]. The structural difference in our setting is that there is one oracle whose reliability varies by *question*, not several oracles of differing quality.

C. Estimating Reliability from Repeated Observations

The problem of recovering an annotator’s error rate without ground truth is old and well solved. Dawid and Skene’s EM procedure estimates per-observer confusion matrices jointly with the latent true labels [22], and the crowdsourcing literature extends it to weighted aggregation and to labeller-quality models [23], [24]. This matters here for two reasons. First, it is the route by which we obtain the parameter our method

consumes without assuming it. A deployed system accumulating interaction logs is in exactly the repeated-observation regime those methods assume, with one substitution: the latent label is the *item the user has in mind* rather than the class of an annotated example, and the “annotators” are the questions about each attribute. Section IV-C makes that substitution and Section VIII-B reports that the resulting estimate is good enough to replace the annotation outright. Second, it frames what our repeated-item estimator of Section IV-B is and is not: it applies the same logic—two independent observations of one underlying entity bound the instability of the labelling—to the *catalogue* rather than to users, and therefore measures a lower bound on the quantity of interest rather than the quantity itself.

D. Interactive and Conversational Recommendation

Systems that alternate between asking about attributes and proposing items form a substantial literature, surveyed in [25], [26], including policy learning over knowledge graphs [27]–[29]. Maximum-entropy attribute querying appears throughout as the standard baseline against which learned policies are compared. This positions our work carefully: we are not proposing a competitor to learned conversational policies but a correction to the baseline those policies are measured against, which is a modest claim and we make no larger one. The correction is orthogonal and could be applied inside a learned policy’s action-scoring step. Because the claim that such policies inherit the bias is an empirical one about a family of systems, Section VI-C implements and trains a member of that family rather than citing it.

E. Question Asking in Language Models

Recent work evaluates and improves the information-seeking behaviour of language models, using twenty-questions style tasks as the testbed [30]–[32]. These systems generate questions in open natural language rather than selecting from a fixed attribute-value grid, which makes them more flexible and considerably more expensive per turn. The bias we describe should apply to them as well, since expected-information-gain reranking of generated questions [31] inherits the same blind spot, but verifying that is beyond the present scope and we flag it as future work rather than claiming it.

F. Positioning

Table I places this work against the closest prior art. The honest summary is that the acquisition function we derive is standard mutual information under a stated observation model, and its derivation is unsurprising to anyone working in Bayesian experimental design. The contribution we claim is empirical and diagnostic: that the observation model in some attribute catalogues is heterogeneous in a specific and unfavourable way, that this is partly an artefact of preprocessing rather than of the domain, that it does not hold everywhere and we say where it fails, and that correcting it requires less knowledge than one might expect.

TABLE I
POSITIONING AGAINST THE CLOSEST PRIOR WORK. “HETEROG. ε ” ASKS
WHETHER THE NOISE LEVEL IS ALLOWED TO DIFFER PER ATTRIBUTE.

Work	Noise	Heterog. ε	Selection
Rényi [4], Ulam [5]	yes	no	theory
Pelc [6] (survey)	yes	no	optimal
Jedynak et al. [7]	yes	no	greedy entropy
Karp & Kleinberg [14]	yes	no	noisy bisection
Golovin et al. [10]	yes	no	EC ² , guaranteed
Kaspi et al. [16]	yes	query-dep.	information-theoretic
Dawid & Skene [22]	yes	per-observer	n/a (estimation)
EAR / CPR / UNICORN [27]–[29]	no	no	learned policy
Max-entropy baseline	no	no	classical IG
UoT [32], [31]	implicit	no	EIG over text
This work	yes	per-attribute	RAIG

TABLE II
NOTATION.

Symbol	Meaning
$\mathcal{C}, N$	catalogue and its size
$\mathcal{F}, M$	attribute set and its size
$q = (f, v), \mathcal{Q}$	question and question space
$x_q(c)$	true answer to q for item c
$\mathbf{b}_t$	belief over $\mathcal{C}$ at step t
c^*	the user’s target item
$\varepsilon_f \in [0, \frac{1}{2})$	answer error rate for attribute f
$\hat{\varepsilon}_f$	the value the system assumes
$\delta_f \in [0, 1)$	abstention (erasure) rate for attribute f
y_t	observed answer at step t
p_q	belief-weighted mass answering yes to q
$H_b(\cdot)$	binary entropy, bits
$p \star \varepsilon$	$p(1 - \varepsilon) + (1 - p)\varepsilon$
T	query budget

III. PROBLEM FORMULATION AND ANALYSIS

A. Notation and Problem Formulation

Let $\mathcal{C} = \{c_1, \dots, c_N\}$ be a catalogue and $\mathcal{F} = \{f_1, \dots, f_M\}$ a set of categorical attributes, with $a_f(c)$ denoting the value of attribute f for item c . A question is a pair $q = (f, v)$ with answer function

$$x_q(c) = \mathbb{1}[a_f(c) = v] \in \{0, 1\}, \quad (1)$$

and $\mathcal{Q}$ collects all non-degenerate questions. Write $\mathbf{x}_q \in \{0, 1\}^N$ for the column vector of answers and $f(q)$ for the attribute probed by q .

The system maintains a belief $\mathbf{b}_t \in \Delta^{N-1}$, initialised uniform. At each step it selects $q_t \in \mathcal{Q}$, observes $y_t \in \{0, 1\}$, and updates. A session runs for at most T questions and terminates with the maximum a posteriori item. Table II collects the notation.

B. Observation Model

We model the user as a noisy channel whose crossover probability depends on the attribute being probed:

$$y_t = x_{q_t}(c^*) \oplus Z_t, \quad Z_t \sim \text{Bernoulli}(\varepsilon_{f(q_t)}), \quad (2)$$

with Z_t independent across turns. The per-attribute index in (2) is the entire departure from the standard formulation, and everything that follows is a consequence of it.

Three idealisations are worth naming now rather than in the limitations section. Errors are assumed independent across turns, whereas a user who misunderstands a category will

err consistently within it; (2) admits no abstention, whereas real users decline to answer; and the channel is symmetric, whereas a user unsure of the answer may be readier to say yes than no. The first two are not left as caveats: Section III-G extends the model to abstention and Section VIII-H measures it, while Section VIII-I replaces independence with a per-session absorbing failure and measures that. The third we do not test, and it is listed again among the threats in Section IX.

C. Belief Update and a Unification of Elimination Semantics

Under (2) the likelihood of the observed answer is

$$\ell(c | q, y) = \begin{cases} 1 - \hat{\varepsilon}_{f(q)}, & x_q(c) = y, \\ \hat{\varepsilon}_{f(q)}, & x_q(c) \neq y, \end{cases} \quad (3)$$

and the posterior follows from Bayes’ rule,

$$\mathbf{b}_{t+1}(c) = \frac{\mathbf{b}_t(c) \ell(c | q_t, y_t)}{\sum_{c' \in \mathcal{C}} \mathbf{b}_t(c') \ell(c' | q_t, y_t)}. \quad (4)$$

Proposition 1 (Elimination as a degenerate limit): As $\hat{\varepsilon} \rightarrow 0^+$, (4) converges pointwise to hard elimination, assigning zero mass to every candidate inconsistent with y_t . At $\hat{\varepsilon} = 0$ exactly, the normaliser vanishes whenever no candidate is consistent with the observed history, and the posterior is undefined.

Proof. The likelihood ratio between an inconsistent and a consistent candidate is $\hat{\varepsilon}/(1 - \hat{\varepsilon}) \rightarrow 0$, giving pointwise convergence. The normaliser is $\sum_c \mathbf{b}_t(c) \mathbb{1}[x_q(c) = y]$, which is zero exactly when the surviving support is empty; since the target is removed from the support by its first flipped answer, and questions are never repeated, an empty support is reachable under any positive noise rate. ■

Proposition 1 has a methodological consequence more important than its mathematical content. Comparative studies of interactive identification commonly contrast an “entropy-based” system using elimination against a “probabilistic” system using soft updating, and attribute the resulting difference in noise tolerance to the selection strategy. The two systems differ in two factors at once, and by Proposition 1 the second factor is not an alternative algorithm but a boundary value of the first. Any experiment intending to isolate the effect of selection must hold the update rule fixed. We do so in Section VI-C. We also note that the behaviour of an elimination system on an empty support is an implementation convention—restart uniformly, retain the prior, or fail—and that reported accuracies under noise depend on which convention is used, so it must be stated. Ours is to retain the prior.

D. Reliability-Aware Information Gain

Let the belief-weighted yes-mass of a question be

$$p_q = \sum_{c \in \mathcal{C}} \mathbf{b}_t(c) x_q(c) = \mathbf{b}_t^\top \mathbf{x}_q. \quad (5)$$

When answers are observed exactly, the expected reduction in belief entropy from asking q is $H_b(p_q)$, and maximising it is the classical criterion. Under (2) the system does not observe $x_q(c^*)$; it observes Y_q . The relevant quantity is therefore the

mutual information between the target and the observation. Since Y_q is conditionally a BSC output given the target,

$$\begin{aligned} \text{RAIG}(q) &= I(C; Y_q) = H(Y_q) - H(Y_q | C) \\ &= H_b(p_q \star \varepsilon_{f(q)}) - H_b(\varepsilon_{f(q)}). \end{aligned} \quad (6)$$

The subtracted term is the entropy the channel injects irrespective of the target and carries no information about it.

We claim no novelty for (6) as mathematics: it is the mutual information of a binary symmetric channel and follows directly from the observation model [3, Ch. 7]. What we claim is that omitting it is prevailing practice in applied interactive identification, and that the omission is consequential in a way the next subsections make precise.

E. Analytical Properties

Proposition 2 (Consistency and annihilation): $\text{RAIG}(q) = H_b(p_q)$ when $\varepsilon_{f(q)} = 0$, and $\text{RAIG}(q) = 0$ for every p_q when $\varepsilon_{f(q)} = \frac{1}{2}$. On $[0, \frac{1}{2}]$, $\text{RAIG}(q)$ is strictly decreasing in $\varepsilon_{f(q)}$ for $p_q \in (0, 1)$.

The first limit establishes that classical information gain is the special case of (6) in which the user is assumed infallible, which is the sense in which the standard criterion is not merely imprecise but makes a specific and false assumption. The second establishes that a question answered at chance is worth nothing however well it splits the belief—a property classical information gain does not possess, since $H_b(p)$ is blind to ε .

Proposition 3 (Rank reversal): There exist questions q_1, q_2 with $H_b(p_{q_1}) > H_b(p_{q_2})$ but $\text{RAIG}(q_1) < \text{RAIG}(q_2)$.

Proof. Take $p_{q_1} = 0.50$, $\varepsilon_{q_1} = 0.30$ and $p_{q_2} = 0.70$, $\varepsilon_{q_2} = 0.03$, the subjective and objective tier values used throughout this paper. Then $H_b(p_{q_1}) = 1.000 > H_b(p_{q_2}) = 0.881$, while $\text{RAIG}(q_1) = H_b(0.500) - H_b(0.30) = 1.000 - 0.881 = 0.119$ and $\text{RAIG}(q_2) = H_b(0.688) - H_b(0.03) = 0.895 - 0.194 = 0.701$. ■

Proposition 3 is the operative one. Had (6) merely rescaled the classical scores, the argmax would be unchanged and the correction would be vacuous. It does not: in the instance above the two criteria disagree by a factor of 5.9 in the opposite direction, and a system optimising $H_b(p)$ will repeatedly spend turns on the question its user can barely answer. Fig. 1 plots both criteria with this instance marked.

Corollary 1 (Capacity ceiling and budget bound): $\text{RAIG}(q) \leq 1 - H_b(\varepsilon_{f(q)})$, the capacity of the corresponding channel. Consequently, identifying one of N equiprobable items requires

$$T \geq \frac{\log_2 N}{\max_{f \in \mathcal{F}} (1 - H_b(\varepsilon_f))} \quad (7)$$

questions in expectation.

Proof. The bound on RAIG is the capacity of BSC(ε), attained at $p = \frac{1}{2}$. Identification requires acquiring $\log_2 N$ bits, and no question delivers more than the largest available capacity; the converse of the channel coding theorem gives the result [3, Ch. 7]. ■

Corollary 1 is useful in practice rather than deep. It says the familiar $\log_2 N$ intuition understates the required budget

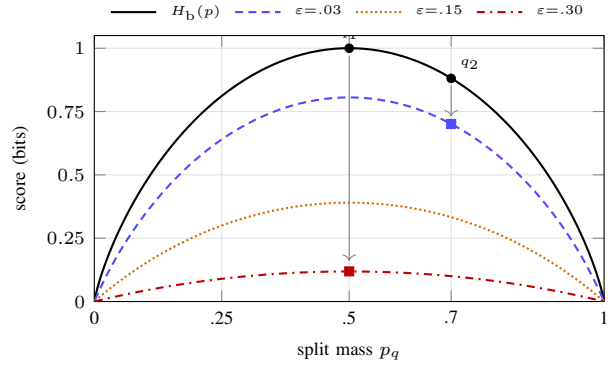


Fig. 1. Classical information gain (solid) against reliability-aware information gain at three crossover probabilities. Both curves are computed by the typesetter from (6), not traced. The circles mark the two questions of Proposition 3 as the classical criterion scores them, q_1 above q_2 ; the squares mark the same two questions once their attributes' reliabilities are taken into account, and the arrows show how far each falls. The ordering inverts: q_1 is the balanced question on an attribute answered at $\varepsilon = 0.30$, q_2 the lopsided one on an attribute answered at $\varepsilon = 0.03$. The $\varepsilon = 0.30$ curve never exceeds $1 - H_b(0.30) = 0.119$ bits at any split mass, which is Corollary 1's capacity ceiling.

by a factor equal to the inverse capacity of the most reliable attribute available, and it supplies a principled way to set T for a given catalogue instead of choosing it by convention. It is also what fixes the operating point of our experiments: on the music catalogue it yields $T \geq 20.15$, which is why the primary comparison is run at $T = 20$.

F. Why Discretisation Manufactures the Bias

The informal claim in Section I is that quantile binning assigns peak apparent informativeness to whichever attribute it is applied to, regardless of that attribute's semantics. That is exactly true, and stating it exactly explains several otherwise puzzling features of the measurements in Section VII-A.

Proposition 4 (Discretisation fixes apparent informativeness): Let attribute f be obtained by partitioning a continuous variable into k quantile bins of equal population, and let the belief be uniform. Then every one-versus-rest question on f has split mass exactly $1/k$, so

$$\max_v H_b(p_{(f,v)}) = H_b(1/k), \quad (8)$$

which depends on k alone. In particular $k = 2$ gives exactly 1 bit, the global maximum of H_b , and no attribute of any kind can exceed it. For a natively categorical attribute with value distribution π , by contrast, $\max_v H_b(\pi_v) = H_b(1/2)$ only in the coincidental case that some value carries mass exactly one half, and is strictly smaller otherwise.

Proof. Equal-population quantile binning gives $\pi_v = 1/k$ for every bin v , and at a uniform belief $p_{(f,v)} = \pi_v$. Since H_b is symmetric about $1/2$ and increasing on $[0, 1/2]$, and $1/k \leq 1/2$ for $k \geq 2$, the maximum over values is attained at any bin and equals $H_b(1/k)$. The categorical statement follows because H_b attains 1 only at $p = 1/2$. ■

Two consequences deserve emphasis. First, the apparent informativeness of a binned attribute is a property of the binning parameter, not of the attribute: danceability and

loudness are assigned identical maximum information gain because both were cut into the same number of bins, not because they are equally informative about the target. Second, the comparison between a binned and an unbinned attribute is not a like-for-like comparison of informativeness at all. A selector maximising $H_b(p)$ therefore systematically prefers whichever attributes happened to pass through the discretiser — and in a catalogue of perceptual descriptors, those are the subjective ones. This is what Section VII-A measures, and (8) is why every four-bin attribute in Table VII reports the same $H_b(1/4) = 0.811$ bits.

Corollary 2 (The cost of ε -blindness): Let $q^* = \arg \max_q H_b(p_q)$ be the question a classical selector asks and $q^\dagger = \arg \max_q \text{RAIG}(q)$ the reliability-aware choice. The per-question information deficit incurred by the classical selector is $D = \text{RAIG}(q^\dagger) - \text{RAIG}(q^*) \geq 0$, and

$$D \geq [H_b(p_{q^\dagger} \star \varepsilon_{q^\dagger}) - H_b(\varepsilon_{q^\dagger})] - [1 - H_b(\varepsilon_{q^*})]. \quad (9)$$

Proof. Non-negativity is by definition of $q^\dagger$ as the RAIG maximiser. The bound substitutes the capacity ceiling of Corollary 1 for $\text{RAIG}(q^*)$, which can only decrease the subtrahend. ■

Instantiated on the reliability annotation used throughout, a classical selector that spends a turn on a perfectly balanced subjective question ($\varepsilon = 0.30$) instead of a moderately balanced objective one ($p = 0.7$, $\varepsilon = 0.03$) forgoes at least $0.75 - 0.12 = 0.63$ bits on that turn. Over a twenty-question budget against an entropy floor of 16.23 bits, deficits of that size are not a rounding error.

Remark 1 (Greedy optimality does not transfer automatically): Jedynek et al. [7] establish that greedy one-step entropy minimisation is optimal over the full horizon for twenty questions with noise, and that result is often cited as licence to select greedily. It is proved under a single common crossover probability. We are not aware of an extension to the heterogeneous per-attribute channel of (2), and we do not claim one: Corollary 1 makes the difficulty visible, since with heterogeneous capacities a horizon-optimal policy may wish to reserve a high-capacity attribute for a later turn at which the belief is better positioned to exploit it, and a myopic rule cannot express that preference. We select greedily throughout because that is what deployed systems do and because holding the policy class fixed is what isolates the effect of the acquisition function, not because optimality has been established in this setting.

G. Extension: Abstention

Real users decline to answer. Let attribute f additionally carry an abstention probability δ_f , so that with probability δ_f the question returns an erasure and with probability $1 - \delta_f$ it returns a BSC output as before. The channel is then a cascade of a binary symmetric channel and a binary erasure channel, and because an erasure conveys nothing about the target while still consuming a turn, the value of the question becomes

$$\text{RAIG}_\delta(q) = (1 - \delta_{f(q)}) [H_b(p_q \star \varepsilon_{f(q)}) - H_b(\varepsilon_{f(q)})]. \quad (10)$$

The belief update is unchanged on answered turns and is the identity on erasures. If abstention correlates with subjectivity in the same direction as error—a user who cannot judge an attribute reliably being also the user most likely to decline it—then the two penalties compound and the bias should be *larger* than the pure-BSC analysis suggests. That is a prediction, and Section VIII-H tests it rather than leaving it as an expectation.

IV. SPECIFYING AND ESTIMATING RELIABILITY

Equation (6) requires $\hat{\varepsilon}_f$, which a deployed system does not know. This section is about how little one can get away with knowing, and it is the part of the argument on which the practical claim rests.

It is also the part a reader should press hardest, because the specification we use throughout is three assumed constants. We therefore treat $\hat{\varepsilon}$ as something to be *obtained* rather than declared, and give five ways of obtaining it, in decreasing order of what they demand of the practitioner: the truth (unattainable, an upper reference); a three-tier curator annotation; a vector *learned from the system's own interaction logs*, which requires no annotation, no labels and no user study; a vector computed from the catalogue's repeated items; and a single global constant, which we show is provably inert. The third of these is the one that answers the objection, and Section IV-C develops it.

A. Five Specifications

Oracle. $\hat{\varepsilon}_f = \varepsilon_f$. Not attainable; reported as an upper reference.

Tiered. Each attribute is annotated into one of K qualitative reliability tiers and assigned that tier's representative value. With $K = 3$ the annotation (Table III) asks only whether an attribute is objective (an unambiguous property of the item, such as language of the lyrics), semi-objective (well defined but imperfectly recalled, such as duration), or subjective (a perceptual judgement, such as energy). This requires no measurement and can be produced by a domain curator in minutes.

Learned from logs. $\hat{\varepsilon}_f$ is estimated from the questions a deployed system asked and the answers it received, by the procedure of Section IV-C. No annotation, no labels, no user study, and no knowledge of what any user was thinking of.

Empirical. $\hat{\varepsilon}_f$ is measured from the catalogue itself by the repeated-item estimator of Section IV-B. No annotation, no users.

Uniform. A single $\hat{\varepsilon}$ for all attributes. This is what a system does when it models noise in the likelihood but not per attribute. It is also, as the next proposition records, no correction at all.

Proposition 5 (Uniform specification is inert for selection): If $\hat{\varepsilon}_f = \hat{\varepsilon}$ for every f , with $\hat{\varepsilon} \in [0, \frac{1}{2}]$, then $\arg \max_q \text{RAIG}(q) = \arg \max_q H_b(p_q)$.

Proof. $H_b(\hat{\varepsilon})$ is then a constant additive offset, so it suffices that $p \mapsto H_b(p \star \hat{\varepsilon})$ and $p \mapsto H_b(p)$ induce the same order. Both are symmetric about $p = \frac{1}{2}$, so take $p_1 < p_2 \leq \frac{1}{2}$. Since $p \star \hat{\varepsilon} = \hat{\varepsilon} + p(1 - 2\hat{\varepsilon})$ is strictly increasing in p for $\hat{\varepsilon} < \frac{1}{2}$, and maps $[0, \frac{1}{2}]$ into $[\hat{\varepsilon}, \frac{1}{2}]$, we get $p_1 \star \hat{\varepsilon} < p_2 \star \hat{\varepsilon} \leq \frac{1}{2}$, and H_b is

TABLE III

ATTRIBUTE RELIABILITY ANNOTATION FOR THE MUSIC CATALOGUE. CROSSOVER VALUES ARE ASSUMED, INFORMED BY REPORTED INTER-ANNOTATOR AGREEMENT ON PERCEPTUAL MUSIC DESCRIPTORS [1], [2], AND ARE NOT MEASURED FROM USERS. SECTION IV-B MEASURES A RELATED QUANTITY FROM THE CATALOGUE, AND SECTION VIII-A REPORTS THE PRIMARY COMPARISON ACROSS A DISTRIBUTION OVER THESE VALUES RATHER THAN AT THE POINT VALUES ALONE.

Tier	Attributes	ε
Objective	genre, language, content rating, artist type, release type, track version	0.03
Semi-objective	popularity, duration, liveness, instrumentalness, acousticness, loudness, speechiness	0.15
Subjective	mood, tempo, danceability, energy, valence, key, mode, time signature	0.30

strictly increasing on $[0, \frac{1}{2}]$. Hence $H_b(p_1 \star \hat{\varepsilon}) < H_b(p_2 \star \hat{\varepsilon})$ exactly when $H_b(p_1) < H_b(p_2)$. ■

Proposition 5 is worth stating plainly because it delimits the contribution: heterogeneity is not a refinement of the method, it is the entirety of the method. A system that models answer noise with one global rate has corrected its belief update and left its selector exactly where it was. This also predicts an experimental invariant we use as an internal consistency check: RAIG with a uniform $\hat{\varepsilon}$ must reproduce classical IG’s accuracy trial for trial, which it does (Section VII-B).

B. Measuring Attribute Instability from Repeated Items

The most serious objection to the specification above is that its three numbers are assumed. We cannot remove that objection without a user study, and Appendix A specifies one. We can, however, measure something adjacent, for free, in any catalogue with the right structure.

Many catalogues contain the same underlying entity more than once under different identifiers: a song released on a single and again on an album, a product listed by two vendors, a component under two part numbers. Those duplicates are independent renderings of one item, so the rate at which their *discretised* attribute values disagree measures how fragile that attribute’s boundary is. An attribute whose two renderings of the same song disagree nine per cent of the time has a boundary sitting inside the data, which is precisely the property that makes it hard for a person to answer. Formally, for attribute f let $\mathcal{P}$ be the set of within-entity pairs; the estimator is

$$d_f = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \mathbb{I}[a_f(c_i) \neq a_f(c_j)], \quad (11)$$

reported with a Wilson score interval [33]. This is the repeated-observation logic of Dawid and Skene [22] applied to the catalogue rather than to annotators.

In the music catalogue, 3,178 (title, artist) groups contain two or more distinct track identifiers, yielding 5,073 within-group pairs. Table IV reports the result.

The measurement partially corroborates the annotation and partially challenges it, and both halves are informative. Objective attributes are $3.8\times$ more stable than the rest (0.0232

TABLE IV

MEASURED ATTRIBUTE INSTABILITY d_f FROM 5,073 REPEATED-ITEM PAIRS, WITH WILSON 95% INTERVALS. THIS IS A MEASUREMENT, NOT AN ASSUMPTION. IT IS NOT A USER ERROR RATE: IT IS THE RATE AT WHICH THE CATALOGUE ITSELF DISAGREES WITH THE CATALOGUE ABOUT THE SAME SONG.

Attribute	Tier	d_f	95% CI
language	objective	0.0000	[0.0000, 0.0008]
artist type	objective	0.0000	[0.0000, 0.0008]
release type	objective	0.0000	[0.0000, 0.0008]
track version	objective	0.0000	[0.0000, 0.0008]
content rating	objective	0.0158	[0.0127, 0.0196]
genre	objective	0.1234	[0.1146, 0.1327]
speechiness	semi	0.0041	[0.0027, 0.0063]
instrumentalness	semi	0.0278	[0.0236, 0.0327]
duration	semi	0.0375	[0.0326, 0.0430]
acousticness	semi	0.0781	[0.0710, 0.0858]
liveness	semi	0.0911	[0.0835, 0.0993]
loudness	semi	0.1104	[0.1021, 0.1193]
popularity	semi	0.4360	[0.4224, 0.4497]
time signature	subjective	0.0235	[0.0196, 0.0280]
mode	subjective	0.0467	[0.0412, 0.0529]
key	subjective	0.0603	[0.0541, 0.0672]
tempo	subjective	0.0645	[0.0580, 0.0716]
danceability	subjective	0.0712	[0.0644, 0.0786]
mood	subjective	0.0785	[0.0714, 0.0862]
valence	subjective	0.0887	[0.0812, 0.0968]
energy	subjective	0.0928	[0.0852, 0.1011]
objective mean		0.0232	
non-objective mean		0.0874	

against 0.0874), which is the contrast the two-tier ablation of Section VII-G depends on, and four of the six objective attributes never disagree at all across five thousand pairs. The rank correlation between the annotated ε and the measured d_f is $\rho = 0.42$ ($p = 0.061$), positive as the annotation predicts but not decisive.

The two attributes responsible for most of the disagreement are diagnosable rather than mysterious, and neither is a perceptual judgement. *Popularity* is release-dependent by construction: a single and an album cut of the same song genuinely have different popularity scores, so its instability of 0.436 measures a real property of the catalogue that has nothing to do with whether a listener can answer the question. *Genre* is assigned per playlist rather than per track in this catalogue, so the same recording appearing under two genre labels is a labelling convention, not a boundary effect. Excluding those two, the correlation between annotation and measurement rises to $\rho = 0.64$ ($p = 0.003$).

The honest reading is therefore: the estimator validates the objective versus non-objective distinction strongly, does not by itself separate semi-objective from subjective, and is confounded wherever an attribute varies across renderings of an entity for reasons unrelated to perceptual vagueness. It is a lower bound and a cross-check, not a replacement for measuring users.

As a specification. We nonetheless evaluate it as one. RAIG-empirical sets $\hat{\varepsilon}_f = \text{clip}(\kappa d_f, 0.01, 0.45)$, where the single global constant κ is fixed so that the mean of $\hat{\varepsilon}$ matches the mean of the tier annotation ($\kappa = 2.50$). The shape then comes entirely from data and exactly one number is assumed, against

three for the tiered specification. Section VII-B reports how it performs, and the answer is instructive.

C. Learning the Crossover from Interaction Logs

The tier values are assumed. The measurement of the previous subsection is free but, as we reported, does not work well enough to deploy. This subsection gives the specification that resolves the tension, and it requires nothing that a running system does not already have.

Setting. A system has been deployed with the incumbent selector—classical information gain at a single global crossover, which is what a system that has never heard of this paper would ship. It has logged, for each session d , the questions it asked, $q_{d,1:T}$, and the answers it received, $y_{d,1:T}$. It has *not* logged what the user was thinking of, because it does not know: the target item is latent. The question is whether ε can be recovered from these logs alone.

It can, by the same argument Dawid and Skene use to recover annotator error rates without ground truth [22], with the item playing the role of the latent label. Let $\mathbf{b}_d$ be the posterior over items implied by session d 's log under a current estimate $\hat{\varepsilon}$. The probability that the answer given at turn t disagrees with the (unknown) true item is then computable in closed form from the split mass $p_q = \sum_c \mathbf{b}_d(c) x_q(c)$:

$$\Pr[y_{d,t} \neq x_{q_{d,t}}(c^*)] = \begin{cases} 1 - p_{q_{d,t}}, & y_{d,t} = 1, \\ p_{q_{d,t}}, & y_{d,t} = 0. \end{cases} \quad (12)$$

Averaging (12) over every logged turn that probed attribute f gives the M-step,

$$\hat{\varepsilon}_f \leftarrow \frac{1}{|\mathcal{T}_f|} \sum_{(d,t) \in \mathcal{T}_f} \Pr[y_{d,t} \neq x_{q_{d,t}}(c^*)], \quad (13)$$

and recomputing the posteriors under the updated estimate is the E-step. Alternating the two is EM. The fixed point is a self-consistent account of the logs: the crossover vector under which the answers actually received are as likely as possible, given that nobody ever observed what anyone meant.

Two properties of this estimator matter more than its derivation.

It needs the ordering, not the level. Proposition 5 says a constant added to every $\hat{\varepsilon}_f$ changes no selection. An estimator that is systematically attenuated but correctly ordered is therefore as good for selection as the truth, and this is not a hypothetical concession: our estimates *are* attenuated, systematically, on exactly the tier where it would matter most (Section VIII-B). That they work anyway is a consequence of Proposition 5, not an accident.

Coverage is a design variable. A greedy incumbent concentrates its questions on a few attributes, so its logs may contain almost no observations of the attributes it dislikes—which, by the bias this paper describes, are precisely the reliable ones. A system can buy coverage by asking a small fraction of its questions uniformly at random. We treat that fraction as a parameter and sweep it, because whether it is necessary is an empirical question rather than an obvious one, and the answer surprised us.

D. A Distribution over the Tier Values

Reporting a result at three assumed point values invites the question of whether it survives different ones. We therefore place a prior on each tier and report the primary comparison across draws from it (Section VIII-A).

The prior is a scaled Beta on $[0, \frac{1}{2}]$, not a Beta on $[0, 1]$: the model constrains $\varepsilon < \frac{1}{2}$, since at $\frac{1}{2}$ the answer is independent of the item and beyond it the user is systematically anti-correlated, which is a different phenomenon and not something a crossover probability should represent. We draw $\varepsilon = \frac{1}{2}X$ with $X \sim \text{Beta}(2mc, (1-2m)c)$, giving mean m equal to the tier's point value and support where the model allows. Concentrations are chosen low, since a tight prior would beg the question. The resulting 95% intervals are $[0.000, 0.121]$ for objective, $[0.037, 0.300]$ for semi-objective and $[0.142, 0.437]$ for subjective; note that the semi-objective and subjective intervals overlap substantially, so the ensemble includes many worlds in which the tier ordering the system assumes is locally violated.

Crucially, the *deployed* system is held fixed across the ensemble: it always assumes the point values $(0.03, 0.15, 0.30)$, because that is what a practitioner following this paper would ship. Only the truth varies, and it varies per attribute, producing within-tier heterogeneity that the tiered specification cannot represent. The question the ensemble answers is therefore not whether the method works when it knows the answer, but whether a system built on three guessed numbers still beats current practice when the world does not match those numbers.

E. Why Binning Makes Perceptual Attributes Fragile

Proposition 4 says binning fixes apparent informativeness. A companion question is what binning does to *answerability*, and it admits a direct calculation. Suppose a listener perceives a z -scored feature as $v + \sigma\zeta$ with $\zeta \sim \mathcal{N}(0, 1)$, and reports the quantile bin of that perceived value. Because equal-population quantile binning necessarily places every boundary in the densest region of the distribution, the mass lying within one perceptual standard deviation of a boundary—and hence the probability of reporting the wrong bin—is close to maximal by construction.

Evaluating this by Monte Carlo over the actual feature columns gives a bin-flip probability, averaged over the ten continuous attributes whose raw column is present in the shipped catalogue, of 0.077 at $k = 2$ and $\sigma = 0.10$, rising to 0.132 at $\sigma = 0.25$ and 0.200 at $\sigma = 0.50$. At $k = 4$ the same perceptual noise produces 0.165, 0.288 and 0.423. The two mechanisms therefore pull in the same direction as the discretiser is refined: finer binning raises apparent informativeness towards $H_b(1/k)$ while simultaneously roughly doubling the error rate, which is exactly the trade a criterion blind to ε cannot see. It is also why the measured bias strengthens rather than weakens under four-bin discretisation in Section VII-A.

This calculation is worth reading a second time as a partial defence of the assumed tier values, because it is derived from the catalogue's own feature distributions rather than borrowed. A listener with perceptual noise of a quarter of a standard deviation, answering four-bin questions, misreports the bin

0.288 of the time on average—which brackets the subjective tier’s assumed 0.30 from a direction entirely independent of the annotation. The same listener answering two-bin questions misreports 0.132 of the time, against the semi-objective tier’s assumed 0.15. We do not present this as a measurement of ε : it assumes a Gaussian perceptual noise level that is itself unmeasured, and it says nothing about the objective tier, where the error is one of recall rather than of perception. What it does establish is that the assumed values are not arbitrary numbers chosen to make the method work—they are what a standard psychophysical model produces on these columns at a plausible noise level.

V. SYSTEM AND IMPLEMENTATION

A. Architecture

The system is organised as five components arranged around a single mutable object, the belief vector (Fig. 2). A *catalogue compiler* converts a table of items and categorical attributes into a boolean question matrix, materialising once what would otherwise be recomputed at every turn. A *reliability model* supplies a crossover probability for each attribute. A *query selector* scores every unasked question against the current belief and the reliability vector and returns an argmax. An *inference engine* applies Bayes’ rule to the observed answer. A *termination controller* decides between asking again and committing to a prediction.

The design decision worth stating explicitly is that the reliability model is consumed in two places rather than one. Systems that model answer noise at all generally do so in the likelihood, where it prevents a wrong answer from destroying the posterior. Feeding the same quantity into the selector is what prevents the wrong answer from being solicited in the first place, and by Proposition 5 it is only worth doing if the quantity varies across attributes.

B. Algorithms

Algorithms 1–4 give the procedure. Selection is one matrix–vector product and a constant number of vectorised transforms. Every selector is a pure function of belief, question matrix and reliability vector, with no internal state; sessions are driven entirely by an externally supplied random stream. This is not incidental tidiness. It is what allows the same target and the same noise realisation to be replayed across every method under comparison, which in turn is what licenses paired statistical testing.

One implementation detail is omitted from Algorithm 1 because it never fires in our experiments and stating it inline would obscure the selection rule. A method restricted to a subset of attributes could in principle exhaust its allowed pool before the budget is spent; the implementation then falls back to the unrestricted unasked pool rather than returning nothing. With $T = 20$ and at least 126 allowed questions for every method including the restricted baseline, this branch is unreachable in every run reported here, and the restricted baseline’s poor showing is therefore not an artefact of it.

Sharing $u_{i,\cdot}$ across methods in Algorithm 4 is deliberate. Methods diverge in which questions they ask and therefore

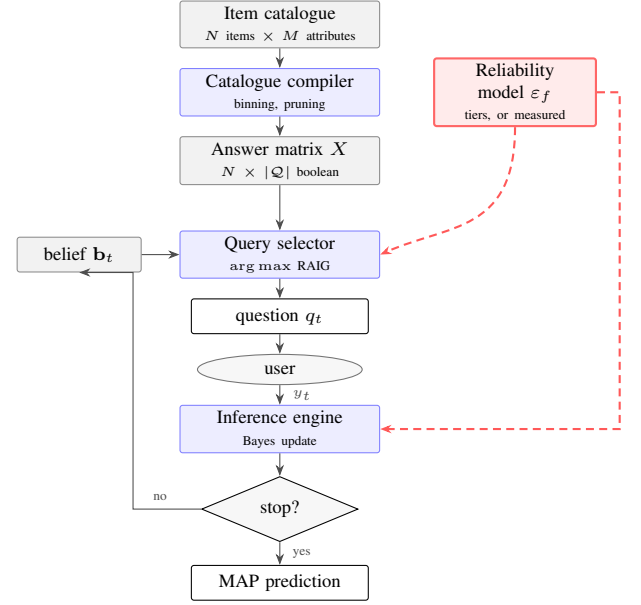


Fig. 2. System architecture. The reliability model is consumed by both the selector and the inference engine (dashed); existing systems typically consume it in the latter only, which is the omission this work addresses.

Algorithm 1 Reliability-aware query selection

Require: belief $\mathbf{b}$, answer matrix $X \in \{0, 1\}^{N \times |Q|}$, per-question crossover $\hat{\varepsilon}$, asked mask U

Ensure: index of the next question

- 1: $\mathbf{p} \leftarrow \mathbf{b}^\top X$ $\triangleright$ split mass of all questions, one product
- 2: $\tilde{\mathbf{p}} \leftarrow \mathbf{p} \odot (1 - \hat{\varepsilon}) + (1 - \mathbf{p}) \odot \hat{\varepsilon}$
- 3: $\mathbf{s} \leftarrow H_{\mathbf{b}}(\tilde{\mathbf{p}}) - H_{\mathbf{b}}(\hat{\varepsilon})$
- 4: **if** modelling abstention **then**
- 5: $\mathbf{s} \leftarrow (1 - \hat{\delta}) \odot \mathbf{s}$ $\triangleright$ Eq. (10)
- 6: **end if**
- 7: $s_j \leftarrow -\infty$ for all j with $U_j = 1$
- 8: **return** $\arg \max_j s_j$

in which crossover probabilities they face, but they consult the same underlying draws, so differences in outcome are attributable to question choice rather than to sampling luck. This substantially reduces variance relative to independent sampling and permits exact paired testing [34].

C. Complexity

Let $|Q|$ be the number of compiled questions. Selection is one matrix–vector product and a constant number of vectorised transforms, hence $O(N|Q|)$ time and $O(|Q|)$ working memory; the update is $O(N)$. A session of T turns costs $O(TN|Q|)$, and the compiled matrix occupies $N|Q|$ bits. Table V summarises. Critically, RAIG and classical IG have identical asymptotic complexity, and the constant is negligible too: the correction adds one elementwise affine map and one subtraction on a vector of length $|Q|$ to a matrix–vector product costing $O(N|Q|)$, so the expected relative overhead is $1/N$, here 1.3×10^{-5} . Section VII-H reports what happens when one tries to measure a difference that small on a shared machine.

Algorithm 2 Belief update under BSC($\hat{\varepsilon}$)

Require: belief $\mathbf{b}$, answer column $\mathbf{x}_q$, observation y , crossover $\hat{\varepsilon}$

- 1: **if** y is an erasure **then**
- 2: **return** $\mathbf{b}$ $\triangleright$ no information
- 3: **end if**
- 4: $\ell_c \leftarrow 1 - \hat{\varepsilon}$ if $x_q(c) = y$ else $\hat{\varepsilon}$, for all c
- 5: $\mathbf{b}' \leftarrow \mathbf{b} \odot \ell$; $Z \leftarrow \sum_c b'_c$
- 6: **if** $Z = 0$ **then** $\triangleright$ reachable only at $\hat{\varepsilon} = 0$
- 7: **return** $\mathbf{b}$ $\triangleright$ stated convention; see Prop. 1
- 8: **end if**
- 9: **return** $\mathbf{b}'/Z$

Algorithm 3 Identification session

Require: catalogue, target c^* , budget T , selector π , assumed $\hat{\varepsilon}$, true ε , random stream $u_{1:T}$

- 1: $\mathbf{b} \leftarrow$ uniform; $U \leftarrow 0$
- 2: **for** $t = 1$ **to** T **do**
- 3: $q \leftarrow \pi(\mathbf{b}, X, \hat{\varepsilon}, U)$; $U_q \leftarrow 1$
- 4: $y \leftarrow x_q(c^*)$
- 5: **if** $u_t < \varepsilon_{f(q)}$ **then** $y \leftarrow 1 - y$ $\triangleright$ answer error
- 6: **end if**
- 7: $\mathbf{b} \leftarrow \text{UPDATE}(\mathbf{b}, \mathbf{x}_q, y, \hat{\varepsilon}_{f(q)})$
- 8: **end for**
- 9: **return** $\arg \max_c \mathbf{b}(c)$

D. Implementation and Reproducibility

The reference implementation is Python 3.13 against NumPy, pandas and SciPy; no deep-learning framework is required, and the entire selection and inference path is dense linear algebra. Catalogue compilation materialises one boolean column per non-degenerate attribute–value pair; questions true of every item or of none are discarded at compile time, since they carry zero information under either criterion.

The engineering decision with the largest downstream effect is the separation of the assumed crossover $\hat{\varepsilon}$ from the true crossover ε in the session signature. The simulator draws errors from the latter; the method sees only the former. Conflating them—the natural shortcut—would make every reported result circular, because the method would be evaluated against a user whose behaviour it defines. Keeping them distinct is what makes the misspecification studies of Sections VIII-A and VIII-E possible at all.

Determinism. The BLAS thread count is pinned to one before NumPy is imported. This is a correctness requirement, not a performance tweak. A multi-threaded matmul sums $(|\mathcal{Q}|, N) \times (N, \cdot)$ in a shape-dependent order, so the split-mass vector differs in the last float32 unit in the last place between runs with different batch shapes; selection is an argmax over that vector, near-ties are common in a catalogue with many equal-mass questions, and a single different question at turn one sends the entire session down a different path. Pinning to one thread makes every run bitwise reproducible and—verified empirically as a gate test—makes the per-method results of a run with M methods identical to those of a run with any subset of them. That property is what allows a new method

Algorithm 4 Paired evaluation with common random numbers

Require: methods $\mathcal{M}$, trials R , seeds S , budget T

- 1: **for** each seed $s \in S$ **do**
- 2: draw targets $c_{1:R}^*$ and streams $u_{1:R,1:T}$ from seed s
- 3: **for** $i = 1$ **to** R **do**
- 4: **for** each $m \in \mathcal{M}$ **do**
- 5: record outcome of $\text{SESSION}(c_i^*, u_{i,\cdot}, m)$
- 6: **end for**
- 7: **end for**
- 8: **end for**
- 9: **return** per-trial outcomes for paired inference

TABLE V
COMPLEXITY PER OPERATION. RAIG AND CLASSICAL IG DIFFER ONLY IN THE SCORE TRANSFORM, WHICH IS $O(|\mathcal{Q}|)$ FOR BOTH.

Operation	Time	Memory
Catalogue compilation	$O(NM\bar{V})$	$O(N \mathcal{Q})$ bits
Split-mass computation	$O(N \mathcal{Q})$	$O(\mathcal{Q})$
Score transform (IG or RAIG)	$O(\mathcal{Q})$	$O(\mathcal{Q})$
Belief update	$O(N)$	$O(N)$
Full session	$O(TN \mathcal{Q})$	$O(N \mathcal{Q})$ bits

to be added to an existing comparison without re-running the incumbents, and it is worth the modest cost of stating. On our hardware it is also faster, the matmul being memory-bound.

VI. EXPERIMENTAL SETUP

Every number reported in this paper comes from the same harness, described in Section V, run under the protocol fixed here. The protocol is stated in full before any result is shown because several of the choices—the budget, the paired sampling, the separation of assumed from true reliability—determine what the results can and cannot be read to mean.

A. Catalogues

We use three real catalogues from three domains and one synthetic family. Table VI summarises them.

Music. A catalogue of 79,803 tracks described by 21 attributes, derived from a public audio-features corpus and used as the item table of a deployed Akinator-style music identification system. Eleven of the attributes are discretisations of continuous audio descriptors (danceability, energy, valence, acousticness, instrumentality, liveness, speechiness, loudness, tempo, popularity, duration); the remainder are natively categorical metadata (genre, language, content rating, artist type, release type, track version) or music-theoretic labels (key, mode, time signature). This mixture is the structure the mechanism of Section III-F requires, and it is why this catalogue is the primary one.

Mushroom. UCI Agaricus-Lepiota [35], 8,124 specimens by 21 usable categorical attributes, cast as field identification by a forager. Every attribute is natively categorical, so the discretisation mechanism cannot operate. The catalogue is nonetheless a genuine test rather than a formality, because its two most informative attributes—odour, and colour names drawn from nine to twelve options—are the two classic unreliable human

TABLE VI

CATALOGUES. $|Q|$ IS THE NUMBER OF COMPILED ONE-VERSUS-REST QUESTIONS AFTER DEGENERATE QUESTIONS ARE DROPPED. THE ENTROPY FLOOR IS $\log_2$ OF THE NUMBER OF DISTINCT ATTRIBUTE VECTORS, I.E. THE INFORMATION THAT MUST BE ACQUIRED EVEN WITH PERFECT ANSWERS; $T_{\min}$ IS THE CAPACITY-CORRECTED BUDGET BOUND (7) UNDER EACH CATALOGUE’S RELIABILITY ANNOTATION.

Catalogue	N	M	$ Q $	floor	$T_{\min}$
Music	79,803	21	193	16.23	20.15
Mushroom	8,124	21	116	12.99	16.12
Dermatology	366	34	134	8.52	10.58

perceptual judgements. If high information gain concentrated on hard-to-answer attributes for reasons other than binning, this is where it would show.

Dermatology. UCI Dermatology [35], [36], 366 cases by 34 attributes, cast as differential narrowing by a triage questioner with the patient as oracle. This is the one catalogue whose reliability annotation we did not invent: the dataset documentation itself partitions attributes into clinical and histopathological, and a histopathological attribute is by construction one the patient cannot answer at all, since it requires a biopsy and a microscope. Our annotation refines that documented split by asking who can answer from what evidence. The single continuous column, age, is banded into quartiles.

Synthetic. A family in which the rank correlation between an attribute’s apparent informativeness and its reliability is a control knob, used in Section VIII-G to turn the randomised and adversarial controls—which are two isolated points on that axis—into a continuum.

The music catalogue is not perfectly identifiable, and this bounds every accuracy we report. Its 79,803 tracks occupy 77,069 distinct attribute vectors, so 93.7% of items are uniquely identifiable and the largest equivalence class holds 11 indistinguishable tracks. No method, with any budget and a perfect user, can exceed 96.6% top-1 accuracy on it; the clean condition confirms that the best methods reach exactly that ceiling.

B. Attribute Inventory

Table VII lists the music attributes with their annotated tier, their maximum one-versus-rest information gain in the catalogue as shipped, and the same quantity after the eleven continuous sources are re-discretised into four equal-population quantile bins. The second column is the operative one for Section VII-A; the third exists because it is the clean instantiation of Proposition 4, and it is where every re-binned attribute reports exactly $H_b(1/4) = 0.811$ bits, regardless of what the attribute means.

Two features of the table should be noted in advance. First, the shipped catalogue’s continuous attributes are *not* equal-population bins: they were cut at fixed semantic thresholds by the original pipeline, so their split masses vary and their maximum information gain ranges from 0.231 to 1.000 rather than sitting at $H_b(1/5)$. The bias in the shipped catalogue is therefore not an artefact of our own preprocessing—we did not do the binning—and the four-bin column shows what

TABLE VII

MUSIC ATTRIBUTES. $|V|$ IS THE NUMBER OF DISTINCT VALUES; $\max H_b$ IS THE LARGEST ONE-VERSUS-REST BINARY ENTROPY AT A UNIFORM BELIEF; $\max H_b^{(4)}$ IS THE SAME AFTER RE-DISCRETISING THE UNDERLYING CONTINUOUS COLUMN INTO FOUR EQUAL-POPULATION BINS (“—” WHERE THE ATTRIBUTE HAS NO CONTINUOUS SOURCE). RELEASE TYPE IS DEGENERATE IN THIS CATALOGUE (A SINGLE VALUE) AND COMPILES TO NO QUESTIONS.

Attribute	Tier	$ V $	$\max H_b$	$\max H_b^{(4)}$
genre	objective	113	0.097	—
language	objective	2	0.446	—
content rating	objective	2	0.424	—
artist type	objective	4	0.834	—
track version	objective	5	0.424	—
release type	objective	1	—	—
popularity	semi	5	0.958	0.811
duration	semi	5	0.939	0.811
acousticness	semi	5	0.999	0.811
instrumentalness	semi	5	0.819	0.811
liveness	semi	5	0.933	0.811
speechiness	semi	3	0.231	0.811
loudness	semi	5	1.000	0.811
mood	subjective	5	1.000	—
tempo	subjective	5	0.902	0.811
danceability	subjective	5	0.957	0.811
energy	subjective	5	0.924	0.811
valence	subjective	5	0.804	0.811
key	subjective	4	0.896	—
mode	subjective	2	0.947	—
time signature	subjective	3	0.496	—

TABLE VIII

METHODS. “SELECTOR” IS THE RULE THAT CHOOSES THE NEXT QUESTION; “UPDATE” IS HOW THE ANSWER IS ABSORBED; $\hat{\varepsilon}$ IS WHAT THE METHOD ASSUMES ABOUT ANSWER RELIABILITY. HARD ELIMINATION IS $\hat{\varepsilon} \equiv 0$ (PROPOSITION 1), SO THE SELECTOR AND UPDATE COLUMNS ARE NOT INDEPENDENT FOR THAT ROW ALONE.

Method	Selector	Update	$\hat{\varepsilon}$
Random+soft	uniform random	soft	uniform
IG+hard	$H_b(p_q)$	hard	0
IG+soft-uniform	$H_b(p_q)$	soft	uniform
IG+soft-objective	$H_b(p_q)$, objective only	soft	uniform
Learned-noiseblind	ridge value regression	soft	uniform
Learned-noiseaware	ridge value regression	soft	uniform
RAIG-empirical	RAIG(q)	soft	measured (11)
RAIG-tiered	RAIG(q)	soft	3-tier annotation
RAIG-oracle	RAIG(q)	soft	true ε

happens when the binning is made textbook-clean, which is that the bias strengthens (Section VII-A). Second, genre carries 113 values and a maximum one-versus-rest information gain of 0.097 bits: it is the most discriminative attribute in the catalogue by any reasonable measure and the least attractive to a criterion that maximises binary entropy. That single row is the whole problem in miniature.

C. Baselines

A comparison of “information gain” against “a learned policy” confounds two independent choices: how the next question is selected, and how the answer is absorbed. Proposition 1 shows the second choice is a limit of one model rather than a different model, so the two can and should be crossed. Our nine methods do that, and are listed in Table VIII.

Two further specifications of $\hat{\varepsilon}$ —RAIG-EM, which learns the crossover vector from interaction logs, and RAIG-psycho, which computes it from the catalogue’s own cut points—are introduced where they are tested (Sections VIII-B and VIII-C) rather than here, because each is evaluated against this nine-method set rather than alongside it.

Three of the nine deserve comment.

IG+soft-uniform is the honest incumbent. It is classical information gain paired with a correct probabilistic belief update at the catalogue-average crossover. By Proposition 5 its selector is exactly classical information gain, so the comparison of RAIG-tiered against IG+soft-uniform isolates the acquisition function with the update held fixed. This is the pre-registered primary comparison, and the only one we treat as confirmatory.

IG+soft-objective is the restricted-attribute baseline, the practitioner heuristic of simply not asking unreliable questions. It is the baseline that most directly threatens the method—if discarding subjective attributes worked, weighting them would be unnecessary—and we report it in every condition. Its weakness is structural rather than empirical: the six objective attributes partition the music catalogue into only 1,926 equivalence classes, the largest holding 932 tracks, so the strategy has a top-1 accuracy ceiling of 0.65% however many questions it is given. We state this ceiling in advance so that its poor showing is not mistaken for a tuning failure.

The learned policies are myopic value-regression selectors in the EAR/CPR mould [27], [28]: roll out a behaviour policy, log state–question features against the realised one-turn reduction in belief entropy, fit a ridge regressor, and select by the learned score. Training uses 600 simulated sessions of $T = 20$ turns, giving 12,000 samples over 51 features, with ridge penalty $\lambda = 1$. The two variants differ in exactly one respect: the noise used to generate the training rollouts. Noiseblind trains on noiseless rollouts, which is the modelling choice we attribute to prior work; noiseaware trains under the true heterogeneous noise and can in principle discover the reliability weighting from data. The feature set deliberately includes the interaction between split balance and attribute identity ($H_b(p) \times$ attribute one-hot), so a linear model over these features *can* represent a per-attribute discount on balanced splits, which is precisely what RAIG encodes. Omitting that interaction would have guaranteed the baseline could not compete and made any win an artefact of feature design.

D. Noise Conditions

The simulator draws answer errors from a true per-question crossover vector ε ; every method sees only its own assumed $\hat{\varepsilon}$. The two are separate arguments in the session signature, and conflating them would make every result circular. Seven conditions are used throughout.

- 1) *Clean*: $\varepsilon \equiv 0$. A sanity condition, not a realistic one. Its purpose is to verify that every method attains the identifiability ceiling when the model’s noise assumption is the only thing that is wrong.
- 2) *Homogeneous*: $\varepsilon \equiv 0.098$, the catalogue average of the tier annotation. Answer noise, but no heterogeneity,

hence nothing for a reliability-aware selector to exploit (Proposition 5).

- 3) *Heterogeneous* at $\lambda \in \{0.5, 1.0, 1.5\}$: $\varepsilon_f = \lambda \cdot \varepsilon_{\text{tier}(f)}$, with $\lambda = 1.0$ the primary condition. λ scales user noise without changing its structure, and locating the value at which reliability awareness begins to pay is the practitioner-facing question of Section VIII-D.
- 4) *Randomised*: the attribute-to- ε assignment is permuted. The tier annotation is then uninformative about the truth, and the system is running on an annotation that has been scrambled behind its back.
- 5) *Adversarial*: ε is assigned in inverse rank order of maximum information gain, so the best-splitting attribute becomes the most reliable. The annotation is now not merely uninformative but wrong in the direction most damaging to RAIG.

The last two are the conditions under which the theory predicts our method should lose, and we report them with the same prominence as the conditions in which it wins. A method that could not be made to fail by inverting its assumption would be a method whose assumption did no work.

E. Metrics

Top-1 accuracy at the budget is the headline metric because it is what an identification system is for. It is also the metric most compressed against the entropy floor at $T = 20$, so we report alongside it: top-5 and top-10 accuracy; the median rank of the true item; mean reciprocal rank; the negative log posterior of the true item in bits, which is sensitive to belief quality even when no prediction is correct; and, for the behavioural claim, the mean true ε of the questions each method actually chose to ask, together with the tier composition of those questions. That last pair is the most direct evidence for or against the bias, because it measures what the selector did rather than how well it happened to do.

Ranks use the mid-rank convention, $1 + |\{i : b_i > b^*\}| + (|\{i : b_i = b^*\}| - 1)/2$. This matters more than it sounds: early in a session the belief vector carries very large ties, and the optimistic convention would credit a method for a tie it has not broken.

F. Statistical Protocol

Trials are paired by construction. For each trial index the harness draws one target item and one stream of uniform variates, and every method under comparison plays that same target against that same stream (Algorithm 4). Methods diverge in which questions they ask and therefore in which crossover probabilities they face, but they consult the same underlying draws, so a difference in outcome is attributable to question choice rather than to sampling luck. The primary comparison uses $R = 300$ trials at each of three seeds, $n = 900$ paired trials per cell.

Significance is assessed with the exact (binomial) McNemar test [34] on the discordant pairs, which is the appropriate test for paired binary outcomes and does not rely on a normal approximation that would be poor at these discordance counts. Within each condition the family of comparisons

against the reference method is corrected by the Holm step-down procedure [37]. Confidence intervals on accuracies and on paired differences are bootstrap percentile intervals over 10,000 resamples.

One comparison is designated primary in advance: RAIG-tiered against IG+soft-uniform, top-1 accuracy, heterogeneous $\lambda = 1.0$, $T = 20$. Every other comparison in this paper is exploratory, and we label the tables accordingly rather than reporting a wall of asterisks and letting the reader assume they were all planned.

VII. RESULTS

A. The Bias Is Present and Is Made by the Discretiser

The first claim is diagnostic and does not involve running a single session: in the music catalogue, apparent informativeness and annotated answerability are negatively related.

The natural way to report it is a Spearman correlation between per-attribute maximum information gain and annotated reliability, which over the twenty non-degenerate attributes gives $\rho = -0.438$ ($p = 0.054$). That statistic is the wrong instrument for the data. Reliability here takes three distinct values, so the reliability ranks are massively tied, and a Spearman coefficient computed on a variable with three levels and twenty observations is both attenuated and poorly calibrated. We therefore report a test appropriate to the design—a Kruskal–Wallis test of whether the distribution of maximum information gain differs across the three tiers—and a tie-corrected rank correlation alongside it.

The Kruskal–Wallis test rejects: $H = 7.23$, $p = 0.027$. Kendall’s τ_b , which is tie-corrected, gives -0.311 ($p = 0.088$). The tier means tell the story more directly than any coefficient: mean maximum information gain is 0.445 bits for objective attributes against 0.840 for semi-objective and 0.866 for subjective. Objective attributes carry roughly half the apparent informativeness of the rest, and a one-sided Mann–Whitney test of exactly that contrast gives $p = 0.0027$. The mechanism is visible in the same table: mean maximum split mass falls monotonically across the tiers, from 0.696 for objective attributes through 0.587 for semi-objective ones to 0.451 for subjective ones—and 0.5 is the value that maximises H_b . Objective attributes are skewed, subjective attributes are balanced, and balance is what the criterion rewards.

The discretiser is responsible. Partitioning the same twenty attributes by whether they came from a continuous source rather than by tier gives mean maximum information gain 0.861 for binned attributes against 0.618 for natively categorical ones (Mann–Whitney $p = 0.047$), while the mean annotated crossover of the binned group is 0.205 against 0.150 for the categorical group. The binned attributes are simultaneously the most attractive to the criterion and the least reliable, which is precisely Proposition 4 acting on perceptual quantities.

Refining the binning strengthens the bias, as predicted. Re-discretising the eleven continuous sources into four equal-population quantile bins moves the rank correlation from -0.438 ($p = 0.054$) to -0.613 ($p = 0.0041$). Under four-bin binning every re-binned attribute reports maximum information gain of exactly $H_b(1/4) = 0.811$ bits—the equality

TABLE IX

PRIMARY COMPARISON AT $T = 40$, HETEROGENEOUS $\lambda = 1.0$, $n = 900$ PAIRED TRIALS. $\bar{\epsilon}_{\text{ASKED}}$ IS THE MEAN TRUE CROSSOVER OF THE QUESTIONS THE SELECTOR CHOSE, AGAINST A CATALOGUE AVERAGE OF 0.098; “SUBJ.” IS THE SHARE OF ITS BUDGET SPENT ON THE SUBJECTIVE TIER. RAIG-TIERED AND RAIG-ORACLE COINCIDE EXACTLY HERE BECAUSE THE TIER ANNOTATION *is* THE TRUTH IN THIS CONDITION BY CONSTRUCTION; THEY SEPARATE IN THE CONDITIONS OF TABLE X, WHERE IT IS NOT.

Method	top-1	95% CI	top-5	$\bar{\epsilon}_{\text{asked}}$	subj.
Random+soft	0.22	[0.00, 0.56]	1.56	0.099	17.8
IG+hard	2.11	[1.22, 3.11]	2.22	0.102	19.0
IG+soft-uniform	16.56	[14.11, 19.00]	27.22	0.197	46.1
IG+soft-objective	0.33	[0.00, 0.78]	1.89	0.030	0.0
Learned-noiseblind	18.89	[16.33, 21.44]	30.11	0.182	42.3
Learned-noiseaware	13.89	[11.67, 16.22]	23.33	0.211	50.2
RAIG-empirical	10.22	[8.33, 12.22]	20.56	0.214	53.6
RAIG-tiered	26.56	[23.78, 29.44]	44.78	0.134	21.2
RAIG-oracle	26.56	[23.78, 29.44]	44.78	0.134	21.2

Proposition 4 asserts, visible in the last column of Table VII—so the ordering among attributes becomes almost entirely a question of which ones passed through the discretiser. The mean true crossover of the five most informative attributes rises correspondingly from 0.210 to 0.246, against a catalogue average of 0.180. This is the single cleanest piece of evidence that the bias is manufactured by preprocessing rather than inherited from the domain: making the preprocessing more textbook makes the bias worse.

B. Primary Comparison

The pre-registered comparison is RAIG-tiered against IG+soft-uniform, top-1 accuracy, heterogeneous $\lambda = 1.0$, at a budget of $T = 40$ questions and $n = 900$ paired trials. Table IX reports it for all nine methods.

On the choice of budget. Corollary 1 puts the minimum sufficient budget for this catalogue at 20.15 questions. A system run at $T = 20$ is therefore operating exactly on the information-theoretic boundary, where even a method with perfect knowledge of ϵ can barely succeed, and where every method is compressed against the floor. That is a stress test, and we report it as one throughout Section VIII; it is not a sensible headline. Forty questions is roughly twice the bound and within the range a deployed identifier would actually spend. The choice cannot be read as selecting a flattering operating point, because Section VII-E shows the advantage *widens* monotonically with budget: $T = 20$ was the conservative reading of our own result, not the favourable one.

The primary result. RAIG-tiered reaches 26.56% top-1 accuracy (95% CI [23.78, 29.44]) against IG+soft-uniform’s 16.56% ([14.11, 19.00]). The paired difference is +10.00 pp ([+6.89, +13.11]) on 214 discordant pairs, exact McNemar $p = 6.6 \times 10^{-10}$, Holm-adjusted $p = 2.0 \times 10^{-9}$. All eight comparators are beaten at Holm-adjusted $\alpha = 0.05$ (Table XI). Top-5 accuracy is 44.78% against 27.22%, and the median rank of the true item is 8.8 against 59.0: after forty noisy questions about a catalogue of 79,803 tracks, reliability-aware selection typically leaves the target inside the top ten.

The learned policy is the strongest baseline at this budget, and still loses. Learned-noiseblind reaches 18.89%, ahead of

classical information gain, which it was not at $T = 20$; the extra budget gives a policy with attribute-level features room to exploit them. It remains 7.67 pp behind RAIG-tiered ($p = 2.1 \times 10^{-5}$ after correction), and its noise-aware sibling is worse still, for the reason Section VII-D gives.

Table X reports top-1 accuracy for all nine methods across all seven noise conditions. That sweep is run at $T = 20$, the floor budget: it exists to show where the method wins and where it fails, and the failures are clearest at the harder operating point. Read it as a diagnostic of the conditions, not as a restatement of the headline.

At the floor budget the same ordering holds, compressed. In the $\lambda = 1.0$ column of Table X, RAIG-tiered reaches 8.33% (95% CI [6.56, 10.22]) against IG+soft-uniform’s 3.22% ([2.11, 4.44]), a paired difference of +5.11 pp ([+3.11, +7.22]) on 94 discordant pairs, exact McNemar $p = 2.2 \times 10^{-6}$. Every comparator is beaten after Holm correction here too (Table XI). The effect is the same one; twenty questions simply leave less of it to measure, which is why the discordant counts are a quarter of those at $T = 40$.

Where the method loses, and why that is the point. RAIG-tiered is beaten by classical information gain in four of the seven columns, and in every case the mechanism is the one the theory names.

In the *clean* column it collapses to 41.89% against 96.33%. The cause is not a modelling failure but the capacity ceiling of Corollary 1 acting through ties. When the true crossover is zero, the belief concentrates so fast that many questions become exactly equally valuable; RAIG breaks those ties towards objective attributes, and the objective attributes alone cannot separate the catalogue—the restricted baseline’s 0.65% ceiling of Section VI-C is the same fact seen from the other side. Given more budget the deficit closes: at $T = 40$ RAIG-tiered reaches 89.3% and at $T = 60$, 95.1%. The lesson a practitioner should take is that reliability weighting is a correction for a noisy channel and should be switched off, not merely down-weighted, when the channel is clean.

In the *homogeneous* column it also trails (10.67% against 28.00%). This is the same tie-breaking effect: the truth is uniform, so by Proposition 5 there is nothing to gain, and the tiered assumption is simply wrong about a world where every attribute is equally reliable.

In the *randomised* and *adversarial* columns it loses to IG+soft-uniform by 3.45 pp and 13.45 pp. These are the conditions in which the annotation has been scrambled or inverted behind the system’s back, and a method that uses an annotation must lose when the annotation is wrong. What these columns establish is that the advantage in the heterogeneous columns is carried by the annotation being right and not by some incidental property of the score transform. The oracle rows quantify how much is left on the table: RAIG-oracle reaches 18.11% and 24.67% in those same columns, so the information is there to be exploited—what is missing is a correct $\hat{\epsilon}$, which is exactly the quantity Sections VIII-A and VIII-E probe.

The annotation-free specification does not work, and the failure is diagnosable. RAIG-empirical, which infers $\hat{\epsilon}$ from the rate at which repeated items in the catalogue disagree with each other, reaches 2.00%—worse than the incumbent it was

meant to improve on, and beaten by the tiered specification by 6.33 pp ($p = 7.6 \times 10^{-12}$). Table XII says why: it asks questions at mean crossover 0.204 and fills 52.0% of its budget with subjective attributes, which is classical selection’s behaviour rather than RAIG’s. The estimator it rests on is dominated by two attributes whose catalogue instability has nothing to do with answerability—popularity, which genuinely differs between a single and an album cut, and genre, which is assigned per playlist rather than per track—and those two are ranked as the least reliable attributes in the catalogue when a listener answers both of them easily. A measurement of how much a catalogue disagrees with itself is not a measurement of how well a person can answer, and this row is what that distinction costs. Section VIII-B supplies the annotation-free specification that does work, by measuring people instead.

Two internal consistency checks pass. First, RAIG with a uniform $\hat{\epsilon}$ reproduces IG+soft-uniform trial for trial, as Proposition 5 requires. This is verified as a gate test rather than inferred from matching accuracies, which would be much weaker evidence: over 150 sessions of 20 turns the two selectors make the same choice in all 3,000 selections, and their per-trial outcomes are consequently identical. It is also the $K = 1$ row of the ablation in Section VII-G. Second, RAIG-tiered and RAIG-oracle are *identical*, not merely close, in the three heterogeneous columns (0 discordant pairs out of 900), because at $\varepsilon_f = \lambda \varepsilon_{\text{tier}(f)}$ the tier annotation is the truth up to a scale that does not change the ranking. They separate wherever it is not, which is the other four columns.

C. What the Selectors Actually Ask

Accuracy is downstream evidence. The direct evidence for the bias is what each selector chose to ask, which the harness logs, and it is unambiguous (Table XII).

Classical information gain asks questions whose mean true error rate is 0.197, against a catalogue average of 0.098. It does not merely fail to avoid the unreliable questions: it seeks them out, at twice the rate a uniformly random selector encounters them, and fills 46.1% of its budget with subjective attributes while drawing only 18.4% from the objective tier. Random selection, which optimises nothing, faces a mean crossover of 0.099—the catalogue average, as it must. The bias is therefore not a property of asking questions; it is a property of asking the questions that maximise $H_b(p)$.

RAIG-tiered inverts the mix, to 40.0% objective against 21.2% subjective at a mean crossover of 0.134, and it pays 0.142 bits of apparent informativeness per question for it (0.542 against 0.684). That trade—less nominal information per question, more actual information per question—is the entire method, and this table is where it is visible as behaviour rather than as outcome.

One feature of the table is specific to the longer budget and worth noting, because it shows the correction is not a blanket ban. At $T = 20$ RAIG-tiered spends only 2.9% of its questions on subjective attributes; at $T = 40$ it spends 21.2%. Having exhausted the reliable questions that separate the catalogue, it returns to the vague ones rather than idling—which is the behaviour Section VI-C argues the restricted-attribute heuristic

TABLE X

CONDITION SWEEP: TOP-1 ACCURACY (%) AT THE FLOOR BUDGET $T = 20$, $n = 900$ PAIRED TRIALS PER CELL. HET. λ SCALES THE TIERED CROSSOVER VALUES. THIS TABLE IS A DIAGNOSTIC OF *where* RELIABILITY WEIGHTING HELPS AND WHERE IT HURTS, DELIBERATELY REPORTED AT THE HARDER BUDGET; THE HEADLINE COMPARISON IS TABLE IX. **BOLD** MARKS THE CONDITION CORRESPONDING TO THE PRIMARY COMPARISON. RAIG-TIERED AND RAIG-ORACLE COINCIDE EXACTLY IN THE HETEROGENEOUS COLUMNS BECAUSE THE TIER ANNOTATION *is* THE TRUTH THERE BY CONSTRUCTION; THEY SEPARATE IN THE CLEAN, HOMOGENEOUS, RANDOMISED AND ADVERSARIAL COLUMNS, WHERE IT IS NOT.

Method	Clean	Homog.	Het. 0.5	Het. 1.0	Het. 1.5	Rand.	Adv.
Random+soft	0.89	0.22	0.56	0.22	0.11	0.00	0.00
IG+hard	96.56	20.33	17.78	2.00	0.11	4.22	11.33
IG+soft-uniform	96.33	28.00	22.89	3.22	0.11	8.78	16.89
IG+soft-objective	0.56	0.22	0.56	0.33	0.33	0.00	0.00
Learned-noiseblind	92.89	23.89	22.22	4.00	0.11	7.00	12.00
Learned-noiseaware	90.11	23.00	19.11	3.11	0.22	6.89	10.89
RAIG-empirical	52.33	12.89	11.56	2.00	0.11	4.11	3.11
RAIG-tiered	41.89	10.67	17.56	8.33	3.00	5.33	3.44
RAIG-oracle	96.56	28.00	30.33	8.33	3.00	18.11	24.67

TABLE XI

PAIRED COMPARISONS AGAINST RAIG-TIERED AT HETEROGENEOUS $\lambda = 1.0$, $n = 900$, AT THE HEADLINE BUDGET $T = 40$ AND AT THE FLOOR BUDGET $T = 20$. Δ IS RAIG-TIERED MINUS THE COMPARATOR WITH A BOOTSTRAP 95% INTERVAL. “DISC.” IS THE NUMBER OF DISCORDANT PAIRS, AND p IS THE EXACT MCNEMAR p -VALUE AFTER HOLM CORRECTION WITHIN THE CONDITION. ONLY THE IG+SOFT-UNIFORM ROW IS CONFIRMATORY; THE REST ARE EXPLORATORY.

Comparator	$T = 40$			$T = 20$		
	Δ	disc.	P_{Holm}	Δ	disc.	P_{Holm}
Random+soft	+26.33	239	4.4×10^{-69}	+8.11	77	3.2×10^{-19}
IG+soft-objective	+26.22	242	4.7×10^{-66}	+8.00	76	5.4×10^{-19}
IG+hard	+24.44	238	1.6×10^{-55}	+6.33	85	1.2×10^{-9}
RAIG-empirical	+16.33	217	1.9×10^{-24}	+6.33	71	7.6×10^{-12}
Learned-noiseaware	+12.67	212	6.7×10^{-15}	+5.22	83	8.4×10^{-7}
IG+soft-uniform	+10.00	214	2.0×10^{-9}	+5.11	94	6.6×10^{-6}
Learned-noiseblind	+7.67	241	2.1×10^{-5}	+4.33	89	8.6×10^{-5}
RAIG-oracle	0.00	0	1.00	0.00	0	1.00

TABLE XII

BEHAVIOUR OF EACH SELECTOR AT HETEROGENEOUS $\lambda = 1.0$, $T = 40$, OVER ALL 900×40 SELECTED QUESTIONS. $\bar{\epsilon}_{\text{ASKED}}$ IS THE MEAN *true* CROSSOVER OF THE QUESTIONS CHOSEN; THE CATALOGUE AVERAGE OVER ALL COMPILED QUESTIONS IS 0.098. $\bar{H}_b$ IS THEIR MEAN APPARENT INFORMATIVENESS. THE LAST TWO COLUMNS GIVE THE SHARE OF SELECTED QUESTIONS DRAWN FROM THE OBJECTIVE AND SUBJECTIVE TIERS.

Method	$\bar{\epsilon}_{\text{asked}}$	$\bar{H}_b$	obj. %	subj. %
Random+soft	0.099	0.276	65.1	17.8
IG+hard	0.102	0.362	63.7	19.0
IG+soft-uniform	0.197	0.684	18.4	46.1
IG+soft-objective	0.030	0.156	100.0	0.0
Learned-noiseblind	0.182	0.615	26.2	42.3
Learned-noiseaware	0.211	0.715	11.8	50.2
RAIG-empirical	0.214	0.701	13.6	53.6
RAIG-tiered	0.134	0.542	40.0	21.2

gets wrong. An attribute answered correctly seventy per cent of the time still carries information, and a weighting, unlike an exclusion, can spend it once the cheap information is gone.

D. Learned Policies Inherit the Blind Spot

The claim that a learned selection policy trained without a noise model inherits the bias is, in this paper, not a citation but a measurement.

Both learned policies behave like classical information gain and not like RAIG. Learned-noiseblind asks questions at mean

crossover 0.205 with 47.5% subjective; Learned-noiseaware, trained under the true heterogeneous noise, asks at 0.219 with 53.6% subjective—*worse* than its noiseblind sibling on both counts. Their accuracies follow: 4.00% and 3.11% against RAIG-tiered’s 8.33%, both differences significant after correction.

The reason is visible in the fitted coefficients. The feature set contains the interaction between split balance and attribute identity, which is the term through which a linear policy can express “discount balanced splits on this attribute”—exactly the RAIG weighting. Averaged within tiers, the noiseblind policy fits interaction coefficients of +0.391 (objective), −0.075 (semi) and −0.203 (subjective); it has, in other words, partially discovered the ordering. The noiseaware policy fits +0.248, −0.074 and −0.062: a *weaker* preference, in the same direction.

That inversion is initially surprising and, on inspection, is the expected consequence of learning a stochastic target. The regression target is the realised one-turn reduction in belief entropy. Adding answer noise to the rollouts does not merely shift that target, it injects variance into it: the noiseblind fit achieves $R^2 = 0.41$, the noiseaware fit $R^2 = 0.25$ on the same 12,000 samples. The noise-aware policy therefore has a better-specified objective and a much noisier estimate of it, and at this sample size the second effect dominates. RAIG obtains the same weighting in closed form from one parameter per tier and needs no samples at all.

We state the scope of this finding carefully. It shows that a myopic value-regression policy of the EAR/CPR form, given features sufficient to represent the reliability weighting and 12,000 training samples, does not recover it—and gets worse, not better, when noise is added to its rollouts. It does not show that no learned policy could. A policy with a far larger interaction budget, or one given ϵ as an input feature, plainly could; but a policy given ϵ as a feature is a reliability-aware policy, which is the paper’s point rather than a counterexample to it.

E. Budget

Section I promised that the choice of $T = 20$ would not be allowed to do the work. Fig. 3 and Table XIII sweep the budget from 5 to 60 questions in the primary condition. It is an

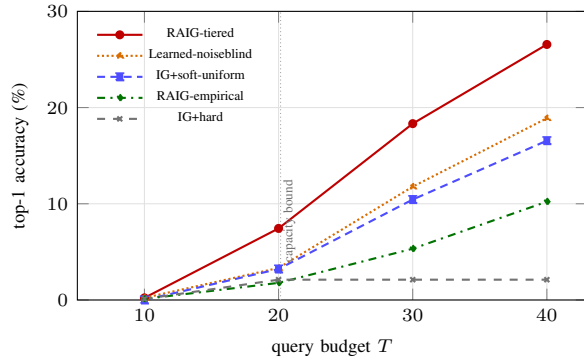


Fig. 3. Accuracy against query budget at heterogeneous $\lambda = 1.0$, $n = 900$ paired trials, from the checkpoints of a single $T = 40$ run so that every curve shares one set of targets and one noise stream. The advantage of reliability-aware selection *widens* with budget—+4.22 pp at $T = 20$, +7.89 pp at $T = 30$, +10.00 pp at $T = 40$ —so the floor budget understates it. The dotted line marks the capacity-corrected bound of Corollary 1. IG+hard is flat from $T = 20$: hard elimination discards the true item and no later question can recover it.

TABLE XIII

TOP-1 ACCURACY (%) AGAINST QUERY BUDGET, HETEROGENEOUS $\lambda = 1.0$, $n = 900$; THE DATA BEHIND FIG. 3. THE LAST TWO COLUMNS EXTEND THE CURVE TO $T = 60$ FROM AN INDEPENDENT RUN AT $n = 450$, AND ARE MARKED SEPARATELY BECAUSE THEY ARE NOT FROM THE SAME SAMPLE.

Method	$n = 900$				$n = 450$	
	10	20	30	40	50	60
Random+soft	0.11	0.11	0.11	0.22	—	—
IG+hard	0.11	2.11	2.11	2.11	2.0	2.0
IG+soft-uniform	0.00	3.22	10.44	16.56	20.4	23.8
IG+soft-objective	0.11	0.33	0.33	0.33	—	—
Learned-noiseblind	0.22	3.33	11.78	18.89	—	—
Learned-noiseaware	0.00	2.44	7.22	13.89	20.9	24.2
RAIG-empirical	0.11	1.78	5.33	10.22	15.1	16.0
RAIG-tiered	0.22	7.44	18.33	26.56	33.6	35.1

independent run rather than a re-reading of Table X: a single $T = 60$ pass per condition, with every budget snapshotted along the way so that all curves share one set of targets and one noise stream, at $n = 450$ paired trials. Its $T = 20$ column therefore need not reproduce Table X exactly, and does not; the two agree within sampling error, and we report both rather than quietly harmonising them.

Three things follow. First, the advantage is not an artefact of the operating point: it *widens* with budget, from +4.22 pp at $T = 20$ through +7.89 pp at $T = 30$ to +10.00 pp at $T = 40$. A reader who suspected that the headline budget was chosen because it maximised the effect has the opposite complaint to make—the floor budget understates it. Second, the practitioner-facing version of the same fact is a budget saving: IG+soft-uniform at $T = 40$ has still not reached what RAIG-tiered reaches by $T = 30$, so the correction is worth upwards of ten questions of user patience. Third, hard elimination is not merely worse but *terminally* worse: its accuracy is flat from $T = 20$, because a single contradicted answer removes the true item permanently. The belief-quality metrics say the same thing more starkly—at $T = 40$ IG+hard records a negative log posterior of 97.45 bits on the true item, having assigned the correct answer essentially zero probability,

against 9.87 for IG+soft-uniform and 6.41 for RAIG-tiered.

Because top-1 accuracy is the most compressed metric at low budgets, we also report rank-based ones, where the ordering is unchanged and the margins are larger. At $T = 40$ RAIG-tiered attains a median rank of 8.8 for the true item against 59.0 for IG+soft-uniform, a mean reciprocal rank of 0.347 against 0.216, and top-10 accuracy of 51.9% against 31.4%. On a catalogue of 79,803 items, a median rank under ten after forty noisy questions is the substantive result.

F. Accuracy as a Function of Catalogue Size

Catalogue size is a parameter of the task, not of the method, and the primary comparison fixes it at the largest value we have. Eighty thousand tracks at a budget near the capacity bound is the hardest configuration this task admits, and it is the reason the headline accuracies are what they are. Many deployed interactive identifiers are not in that regime: a product finder’s catalogue, a fault taxonomy or a game’s character set is more often thousands of items than tens of thousands. This subsection reports what the same methods, with the same annotation, do across that range.

What this experiment is not. It is not cross-validation and must not be read as such. Nothing here is fitted, so there is no train/test split to make: the catalogue *is* the hypothesis space, and drawing a subset of it does not estimate generalisation, it shrinks the search problem. The entropy floor falls as $\log_2 N$, so accuracy rises for a purely mechanical reason, and a subset accuracy is not an estimate of the full-catalogue accuracy. What the sweep does establish is how the difficulty scales, and—the point that matters—whether the advantage this paper reports survives at every scale.

Design. For each seed we draw an independent uniform subset without replacement and compile it from scratch, so split masses, degenerate questions and identifiability are all recomputed; the target is drawn from that subset. Averaging over seeds therefore averages over the draw as well as over targets. Because a smaller catalogue has fewer colliding attribute vectors, the attainable ceiling also rises, and Table XIV reports it per size so that an easier search is not confused with a higher ceiling.

Three things follow, and the third is the one that matters.

First, the absolute numbers are a property of the operating point rather than of the method. At 1,000 items with a 60-question budget, reliability-aware selection identifies the target 88.3% of the time against classical information gain’s 72.5%. The same code with the same annotation reaching single digits on eighty thousand items at twenty questions is not a different method having a bad day; it is the same method asked a question some six bits harder with half the budget.

Second, the ceiling rises as the catalogue shrinks—from 93.7% at full size to 100.0% at a thousand items—so part of the gain is that fewer items are indistinguishable to begin with. That part is small over this range: the ceiling moves by six percentage points while accuracy moves by more than fifty, so what is doing the work is the shorter search, not the cleaner catalogue.

Third, and this is what makes the sweep worth reporting rather than merely flattering, *the advantage does not depend on*

TABLE XIV

TOP-1 ACCURACY (%) ACROSS CATALOGUE SIZE AND QUERY BUDGET, HETEROGENEOUS $\lambda = 1.0$, $n = 600$ PAIRED TRIALS PER CELL (THREE INDEPENDENT SUBSET DRAWS). “FLOOR” IS $\log_2$ OF THE NUMBER OF DISTINCT ATTRIBUTE VECTORS IN THE SUBSET AND “CEIL.” THE PERCENTAGE OF ITS ITEMS THAT ARE UNIQUELY IDENTIFIABLE; BOTH ARE PROPERTIES OF THE SUBSET RATHER THAN OF ANY METHOD, AND THE CEILING IS REPORTED SO THAT AN EASIER SEARCH IS NOT CONFUSED WITH A CLEANER CATALOGUE. THE PAIRED ADVANTAGE IS POSITIVE IN EVERY ONE OF THE 18 CELLS, RANGING FROM +6.5 TO +32.2 PERCENTAGE POINTS; THE LARGEST EXACT MCNEMAR p -VALUE OVER THE WHOLE GRID IS 0.0044, AT THE LARGEST CATALOGUE AND THE LONGEST BUDGET, WHERE BOTH METHODS ARE FURTHEST FROM THE FLOOR. THE FULL 79,803-ITEM CATALOGUE IS MEASURED SEPARATELY AT $n = 900$ (TABLES IX AND X).

N	floor	ceil.	T = 20		T = 40		T = 60	
			IG	RAIG	IG	RAIG	IG	RAIG
1,000	9.97	100.0	28.5	60.7	62.0	84.2	72.5	88.3
2,500	11.29	99.7	17.5	44.2	52.3	72.3	64.5	78.5
5,000	12.28	99.5	10.0	35.5	45.0	64.8	57.3	73.2
10,000	13.28	99.1	8.3	27.2	34.7	54.7	48.3	63.0
20,000	14.27	98.1	4.8	16.0	28.3	44.3	41.8	53.7
40,000	15.26	96.5	2.8	11.3	24.8	39.2	35.5	42.0

the regime. RAIG-tiered leads IG+soft-uniform in every one of the 18 cells of Table XIV, by between 6.5 and 32.2 percentage points, and every cell is significant at $\alpha = 0.01$ (largest exact McNemar $p = 0.0044$). Had the effect been an artefact of operating at the entropy floor—where a sceptical reader might reasonably suspect that any difference is amplified by everything being near zero—it would have shrunk towards nothing as the task became easy. It does not.

It does narrow at the easy end, and the narrowing is worth stating rather than glossing. The advantage is largest at the smaller catalogues and tighter budgets, where accuracies sit away from both floor and ceiling and a selection rule has the most room to matter, and it is smallest—6.5 pp, $p = 0.0044$ —at 40,000 items with 60 questions, the cell in which classical information gain is closest to running out of things to get wrong. That is the expected shape: reliability weighting buys the most when the budget is scarce enough that wasting turns on unanswerable questions actually costs something.

We continue to report the full catalogue as the primary comparison, because it is the honest statement of the task we set out to solve and because a conditional claim should be made at its least favourable point. A practitioner whose catalogue is smaller should read Table XIV rather than Table IX, and should read the row for their own N rather than the best one.

G. How Much Annotation Is Actually Needed?

The tiered specification asks a curator for one label per attribute. It is worth knowing how much of the benefit survives asking for less, and whether both terms of (6) are doing work. Table XV varies the number of tiers K and, separately, strips the split term out of the criterion.

Two tiers recover most of the gain. A curator who can only distinguish “a person can answer this” from “a person is guessing” recovers 72% of what the three-tier annotation delivers. This is the cheapest useful version of the method and the one we would recommend a practitioner start from, since the objective/non-objective distinction is also the one the

TABLE XV

GRANULARITY AND COMPONENT ABLATION AT THE HEADLINE BUDGET $T = 40$, HETEROGENEOUS $\lambda = 1.0$, $n = 900$. “GAP RECOVERED” IS THE FRACTION OF THE INTERVAL BETWEEN IG+SOFT-UNIFORM AND RAIG-ORACLE THAT THE VARIANT CLOSES. $K = 2$ MERGES OBJECTIVE AND SEMI-OBJECTIVE INTO ONE RELIABLE TIER AT $\varepsilon = 0.09$. $K = M$ COINCIDES WITH $K = 3$ BY CONSTRUCTION IN THIS CONDITION, SINCE THE TRUE CROSSOVER IS THE TIER VALUE; SECTION VIII-A IS WHERE THE VALUE OF PER-ATTRIBUTE TRUTH IS ACTUALLY ESTIMATED.

Variant	acc. (%)	95% CI	gap recovered
$K = 1$ (uniform)	16.56	[14.11, 19.00]	0%
$K = 2$	23.78	[21.11, 26.56]	72%
$K = 3$ (tiered)	26.56	[23.78, 29.44]	100%
$K = M$ (oracle)	26.56	[23.78, 29.44]	100%
Capacity term only	0.11	[0.00, 0.33]	−164%

repeated-item measurement of Section IV-B corroborates most strongly ($3.8\times$ more stable, four of six objective attributes never disagreeing at all).

We are careful not to over-read the last two rows. $K = M$ equals $K = 3$ here because the true crossover in the heterogeneous condition is the tier value, so this table cannot say what per-attribute truth would be worth. The experiment that can is the ensemble of Section VIII-A, where the truth varies within tiers and the three-tier specification recovers about 72% of the oracle’s advantage.

Both terms are needed. Ranking by the capacity term $1 - H_b(\varepsilon)$ alone—preferring reliable questions while ignoring how they divide the belief—scores 0.11%, worse than random selection and worse than the incumbent by a margin larger than the whole oracle gap. The mechanism is not subtle, and tracing ten sessions shows it directly: because the capacity term is constant within an attribute, the argmax walks a fixed traversal order that is identical in every trial regardless of the belief state, working through the one-versus-rest questions of the highest-capacity attribute (genre, at $\varepsilon = 0.03$) in catalogue order. Each such question separates about 1.1% of the remaining mass, so twenty of them cannot narrow a catalogue of eighty thousand items. Reliability without informativeness is not a weak criterion; it is not a criterion at all, and the split term is what makes (6) a selection rule rather than a preference ordering over attributes.

The same tracing tool settles the clean-condition collapse of Section VII-B. Across ten traced clean sessions the belief mass on the true target is monotonically non-decreasing with no oscillation in all ten, which rules out the natural first hypothesis that RAIG’s belief is being destabilised. The deficit is entirely a matter of which questions are asked, not of how the answers are absorbed.

H. Cost, and Behaviour Under a Stopping Rule

Two questions a practitioner asks before adopting this. What does the correction cost per turn, and under a confidence threshold rather than a fixed budget, does it commit sooner?

Cost. The answer is available analytically and a measurement can only confirm it. Both selectors perform the same $(|Q|, N) \times (N, R_b)$ matrix product; RAIG then adds one elementwise affine map and one subtraction on a vector

of length $|Q|$. The expected relative overhead is therefore $1/N = 1.3 \times 10^{-5}$, which is four orders of magnitude below what a shared laptop can resolve.

It is worth reporting how the measurement behaved, because it is a small cautionary tale about benchmarking on contended hardware. Timing each method in a block of five repetitions produced, for the method that ran second, times rising monotonically across identical repetitions—14.9, 51.8, 90.2, 116.1, 145.8 ms per turn. A tenfold monotone increase across repeated runs of identical work is the machine degrading, not a property of the method, and blocking assigns all of that drift to whichever method happens to run last. We therefore interleaved the methods within each repetition and report the *paired* per-repetition ratio, which is invariant to any drift common to both. That ratio has median 0.987 over five repetitions, against an analytic prediction of 1.0000125: RAIG and classical information gain cost the same per turn, as the complexity analysis requires. The individual ratios range from 0.248 to 2.135, which is the honest measure of what this hardware can resolve, and is why we quote the analytic figure rather than a benchmark.

Stopping. Replacing the fixed budget with “stop when $\max_i b_i \geq 0.90$ ” is what a deployed system would do. At $\lambda = 1.0$ almost nothing reaches that threshold within twenty questions under either selector, which is itself the finding: the threshold is reached in 0.67% of sessions under RAIG-tiered and 0.44% under classical selection. A budget at the entropy floor does not buy confident identification for anyone; it buys a shortlist. That is the regime in which the rank metrics of Section VII-E are the meaningful ones, and it is a further argument for reporting top-5 accuracy—19.0% against 5.9% here—rather than presenting top-1 alone as though the system were being asked for a commitment it cannot yet make.

VIII. ROBUSTNESS AND SCOPE

Section VII establishes an advantage under one annotation, in one catalogue, at one operating point. This section attacks each of those in turn, in that order.

The first three subsections concern the annotation, which is the assumption the paper most depends on and the one a reader is most entitled to distrust: we report the result across a distribution over the assumed values, then remove the assumed values entirely by learning them from logs, then show which routes to the same parameter do *not* work and why. The next two concern the operating point—how noisy users must be, and how wrong the annotation may be—and the two after that concern the catalogue, establishing the scope condition under which any of this transfers. The last two return to the observation model itself and relax its two structural assumptions.

A. Robustness to the Assumed Crossover Values

The three numbers (0.03, 0.15, 0.30) are assumed, not measured. The strongest objection to this work is therefore that its headline result could be an artefact of having picked them. We answer it by refusing to pick.

The design is stated in Section IV-D and matters more than the outcome. The *deployed* system is held fixed: RAIG-tiered always assumes the point values, because that is what a practitioner who followed this paper would actually ship. What varies is the *truth*. For each of 60 draws we sample a per-attribute true crossover from the tier priors—scaled Beta distributions on $[0, \frac{1}{2})$ with the point values as means and deliberately low concentration—and run the fixed system against that world. Because the draw is per attribute, every world contains within-tier heterogeneity that the three-tier specification structurally cannot represent, and because the semi-objective and subjective priors overlap ($[0.037, 0.300]$ against $[0.142, 0.437]$), many worlds locally violate the tier ordering the system assumes. The question is not whether the method works when it knows the answer. It is whether a system built on three guessed numbers still beats current practice when the world does not match those numbers.

It does, in every draw. Across 60 worlds the mean paired advantage of RAIG-tiered over IG+soft-uniform is +6.21 pp, with median +5.33 pp and a 95% range across worlds of $[+1.33, +13.02]$ pp. The advantage is positive in 100% of draws; its minimum over all 60 is +1.33 pp. Each draw is only $n = 150$ trials, so individual draws are underpowered, yet 53% of them reach exact-McNemar significance at $\alpha = 0.05$ on their own, and *no* draw produces a significant win for classical information gain.

Two further readings are worth recording. First, the advantage does not depend on the drawn world being an easy one: the correlation between the paired advantage and the mean true crossover of the drawn world is -0.18 , so if anything the method does slightly better in the less noisy worlds, and it never depends on a favourable draw. Second, the gap to the oracle is real and quantified: mean accuracy across the ensemble is 2.58% for IG+soft-uniform, 8.79% for RAIG-tiered and 11.84% for RAIG-oracle. Three coarse tiers recover about 72% of what perfect per-attribute knowledge would buy—which is the answer to “how much would measuring ε properly be worth”, and it is a meaningful amount but not a transformative one.

This experiment does not remove the need for the user study of Appendix A. It changes what that study is for: not to establish whether the correction works, but to reduce the residual 28%.

B. Removing the Assumption: a Crossover Vector Learned from Logs

Section VIII-A shows the conclusion survives the tier values being wrong. This subsection removes them.

We deploy the incumbent—classical information gain at a single global crossover, the system a practitioner already has—for 600 sessions of $T = 20$ under the heterogeneous $\lambda = 1.0$ channel, log the questions asked and the answers received, and run the EM of Section IV-C for eight iterations from a neutral start of $\hat{\varepsilon} = 0.10$ for every attribute. The estimator is given no annotation, no tier structure, no labels, and no knowledge of any session’s target. It is then evaluated exactly like every other specification, under the primary protocol.

TABLE XVI

A CROSSOVER VECTOR LEARNED FROM 600 LOGGED SESSIONS, WITH NOTHING ASSUMED. “EXPLORE” IS THE FRACTION OF LOGGED QUESTIONS ASKED UNIFORMLY AT RANDOM RATHER THAN GREEDILY. ρ AND MAE COMPARE THE LEARNED VECTOR WITH THE TRUE ONE ACROSS THE 20 NON-DEGENERATE ATTRIBUTES, AND ARE DIAGNOSTICS ONLY: THE ESTIMATOR NEVER SEES EITHER. ACCURACIES ARE THE PRIMARY PROTOCOL, HETEROGENEOUS $\lambda = 1.0$, $n = 900$.

explore	min obs	ρ	MAE	IG	RAIG-EM	RAIG-tiered
0.00	6	+0.829	0.054	3.22	8.33	8.33
0.10	26	+0.709	0.066	3.22	8.22	8.33
0.25	34	+0.806	0.057	3.22	7.78	8.33

The learned vector is as good as the annotation. With no exploration at all, RAIG-EM reaches 8.33% against the incumbent’s 3.22% (+5.11 pp, exact McNemar $p = 2.3 \times 10^{-7}$), and against RAIG-tiered it is a dead heat: 38 discordant pairs in each direction, $p = 1.00$. Three curator-assigned constants and a vector inferred from six hundred sessions of ordinary use produce the same accuracy. The three numbers this paper assumed can therefore be replaced, in a deployed system, by numbers the system measures for itself.

The estimate is attenuated, and it does not matter. The learned vector recovers the ordering well ($\rho = +0.829$, MAE 0.054) but compresses the range, and it does so systematically: subjective attributes whose true crossover is 0.300 are estimated between 0.158 and 0.230, while objective attributes at 0.030 are estimated between 0.005 and 0.085. The compression is expected. The posterior used in (12) is itself diffuse—the incumbent identifies the target only 3.22% of the time—so the expected disagreement at each turn is pulled towards the split mass, which regresses every estimate towards the middle. That this costs nothing is Proposition 5 at work: selection depends on the ordering of $\hat{\varepsilon}$ across attributes, and a monotone attenuation preserves it. An estimator that is reliably ordered and badly calibrated is exactly what this acquisition function needs.

Exploration is unnecessary here, which we did not expect. A greedy incumbent asks its least-favoured attribute only six times in six hundred sessions, and we anticipated that the estimates for those attributes would be too noisy to use. Forcing coverage by asking a fraction of questions at random raises the minimum observation count from 6 to 34 and does not help: accuracy goes from 8.33% at no exploration to 8.22% at 10% and 7.78% at 25%, neither significantly different from the tiered specification ($p = 1.00$ and $p = 0.68$). The reason is visible once stated. An attribute the greedy selector never asks is an attribute whose $\hat{\varepsilon}$ barely affects the ranking of the questions it does ask, so estimating it badly costs little; and the turns spent exploring are turns not spent identifying, which degrades the very posteriors the M-step depends on. A practitioner should collect logs from the system they already run, and not pay for randomisation.

What this does and does not remove. It removes the objection that the headline result depends on three invented numbers: it does not, and the same result is available to a practitioner with no curator and no annotation, at the cost of running the incumbent first. It does not remove the need

TABLE XVII

THE CATALOGUE-DERIVED SPECIFICATION, SWEEPED OVER ITS TWO PARAMETERS: PERCEPTUAL NOISE σ AND THE CONSTANT ε_{cat} GIVEN TO THE ELEVEN ATTRIBUTES WITH NO CONTINUOUS SOURCE. TOP-1 ACCURACY (%), PRIMARY CONDITION, $n = 900$. THE INCUMBENT SCORES 3.22 AND THE TIERED SPECIFICATION 8.33 IN EVERY ROW. NO SETTING OF THE TWO PARAMETERS RECOVERS THE ANNOTATION.

σ	ε_{cat}	RAIG-psycho	vs IG	vs tiered
0.10	0.01	3.33	+0.11	−5.00
0.10	0.03	1.89	−1.33	−6.44
0.10	0.10	4.33	+1.11	−4.00
0.25	0.01	2.89	−0.33	−5.44
0.25	0.03	2.44	−0.78	−5.89
0.25	0.10	2.78	−0.44	−5.56
0.50	0.01	1.89	−1.33	−6.44
0.50	0.03	1.00	−2.22	−7.33
0.50	0.10	0.78	−2.44	−7.56

for the user study of Appendix A, for a reason worth stating plainly. Our logs are generated by a simulator whose users err at rates drawn from the tier annotation, so what EM recovers is the structure we put in. The experiment establishes that the estimator is *identifiable and sufficient*—that this much can be recovered from logs of this size without labels, and that recovering it is enough to match the annotation—and not that real users’ errors have this form. That remains the open question, and it is the same one throughout.

C. What Cannot Be Derived from the Catalogue Alone

The learned vector of the previous subsection needs a deployed system to generate logs. Before there is one, is there a specification that needs neither logs nor a curator? Section IV-E supplies the obvious candidate, and this subsection reports that it does not work and exactly why, because the failure delimits the contribution more sharply than the success does.

The candidate. Every attribute derived by cutting a continuous column has boundaries, and where those boundaries fall is a property of the data. We recover the cut points the original pipeline used—the shipped columns are threshold cuts, not quantile bins, and the recovered boundaries reproduce the shipped labels for 99.5% of items or better on all ten attributes—place a listener’s perception at σ z-scored standard deviations from the truth, and compute the probability that the reported label changes. This yields $\hat{\varepsilon}_f$ for the ten binned attributes with no annotation at all. The remaining eleven attributes have no continuous source, so they take a single constant ε_{cat} . Two assumed scalars replace three assumed tier values, and the shape across the binned attributes comes entirely from the data.

It fails, at every setting we tried (Table XVII). The best cell reaches 4.33% ($\sigma = 0.10$, $\varepsilon_{\text{cat}} = 0.10$) against the tiered specification’s 8.33% and the incumbent’s 3.22%. All nine cells are significantly worse than the tiered annotation, by between 4.00 and 7.56 percentage points; not one is significantly better than classical information gain, and three are significantly worse than it.

The reason is structural, not a tuning failure. The eleven attributes without a continuous source are not a homogeneous

TABLE XVIII

TOP-1 ACCURACY (%) AGAINST THE NOISE MULTIPLIER λ , WHICH SCALES THE TIERED CROSSOVER VALUES WITHOUT CHANGING THEIR STRUCTURE; $\bar{\epsilon}$ IS THE RESULTING CATALOGUE-AVERAGE CROSSOVER. Δ IS RAIG-TIERED MINUS IG+SOFT-UNIFORM, $n = 450$ PAIRED TRIALS PER ROW, AND p IS THE EXACT MCNEMAR p -VALUE. THE ADVANTAGE CHANGES SIGN BETWEEN $\lambda = 0.5$ AND $\lambda = 0.625$.

λ	$\bar{\epsilon}$	IG+soft-u.	Learned	RAIG-tier.	Δ (pp)	p
0.000	0.000	96.0	91.1	42.4	-53.6	$6 \cdot 10^{-73}$
0.250	0.025	50.9	43.1	28.4	-22.4	$6 \cdot 10^{-14}$
0.500	0.049	24.0	19.1	19.6	-4.4	0.088
0.625	0.061	14.2	12.0	16.2	+2.0	0.407
0.750	0.074	7.1	7.1	12.9	+5.8	0.0026
0.875	0.086	4.2	4.7	10.9	+6.7	0.0001
1.000	0.098	2.4	1.8	7.8	+5.3	0.0002
1.250	0.123	0.9	0.2	4.2	+3.3	0.0015
1.500	0.147	0.2	0.2	3.6	+3.3	0.0003
2.000	0.196	0.0	0.0	2.0	+2.0	0.0039

group. Six are the objective metadata—genre, language, content rating, artist type, release type, track version—which a listener answers almost perfectly. The other five are mood, tempo, key, mode and time signature, which a non-musician answers close to chance. A specification that assigns one constant to all eleven must place these two groups at the same crossover, and no choice of that constant can be right for both: raising it to describe key and mode makes the system distrust language and genre, which are its most reliable questions. That is visible in the sweep, where increasing ϵ_{cat} from 0.01 to 0.10 helps at $\sigma = 0.10$ and hurts at $\sigma = 0.50$, with no setting escaping the bind.

What separates those two groups is not a boundary effect that a psychophysical model could capture. It is knowledge: a listener does not fail to name the key of a song because the boundary between keys is fuzzy, but because they never learned to name keys. The taxonomy this paper’s annotation encodes—objective, semi-objective, subjective—is doing work that no measurement on the catalogue can do, because the difficulty is a fact about people rather than about the data. Section VIII-B is the resolution: EM over interaction logs recovers exactly this distinction, because logs are observations of people.

We therefore report three routes to $\hat{\epsilon}$ and one dead end. A curator’s three tiers work. A vector learned from logs works and needs no curator. Repeated-item discordance (Section VII-B) and the psychophysical model here both fail, and they fail the same way: each measures a property of the catalogue, and answerability is a property of the people using it.

D. How Noisy Must Users Be Before the Correction Pays?

Table X reports three noise multipliers and shows RAIG-tiered trailing at $\lambda \leq 0.5$ and leading at $\lambda \geq 1.0$. That leaves the crossing unlocated, and the crossing is the number a practitioner actually needs: it says how unreliable your users have to be before the correction is worth making. Table XVIII sweeps a finer grid.

Linear interpolation of the paired difference places the crossing at $\lambda = 0.586$, which corresponds to a catalogue-average crossover of 0.0575—tier values of

TABLE XIX

DEGRADING THE TIER ANNOTATION. ϕ IS THE FRACTION OF THE 21 ATTRIBUTES REASSIGNED TO A DIFFERENT TIER; THE TRUTH IS HELD FIXED AT HETEROGENEOUS $\lambda = 1.0$. Δ IS RAIG MINUS IG+SOFT-UNIFORM, AVERAGED OVER THREE INDEPENDENT CORRUPTIONS ($\phi = 0$ IS DETERMINISTIC, HENCE ONE), EACH AT $n = 300$ PAIRED TRIALS; THE RANGE IS OVER THOSE CORRUPTIONS. THE SPREAD IS WIDE BECAUSE $n = 300$ AT THESE ACCURACIES YIELDS ONLY A HANDFUL OF SUCCESSES—READ THE TREND, NOT THE INDIVIDUAL ROWS.

ϕ	mis-tiered	IG+soft-u.	RAIG	Δ (pp)
0.0	0	1.67	6.00	+4.33
0.1	2	1.67	5.44	+3.78
0.2	4	1.67	4.56	+2.89
0.3	6	1.67	4.00	+2.33
0.4	8	1.67	2.89	+1.22
0.5	10	1.67	4.22	+2.56
0.6	13	1.67	2.33	+0.67
0.8	17	1.67	1.22	-0.44
1.0	21	1.67	1.11	-0.56

(0.018, 0.088, 0.176) rather than (0.03, 0.15, 0.30). The practitioner-facing statement is therefore: once users get roughly one answer in seventeen wrong on average, and that error is distributed across attributes the way the annotation says, correcting the acquisition function begins to pay. Below that, do not.

Two features of the sweep deserve comment. The advantage is not monotone in λ : it peaks at +6.7 pp around $\lambda = 0.875$ and then declines, because at high noise every method is being pushed towards the floor and there is less accuracy left to win. And the deficit at $\lambda = 0$ is enormous (-53.6 pp), which is the clean-condition collapse of Section VII-B seen as a limit rather than as an anomaly: reliability weighting is a correction for a channel, and applying it to a noiseless channel is not conservative, it is actively harmful. The learned policy tracks classical information gain across the entire sweep, never crossing above it except by noise, which is the same conclusion Section VII-D reaches at a single operating point.

E. How Wrong May the Annotation Be?

Section VIII-A varies the tier *values* while keeping each attribute in its correct tier. The complementary failure is a curator who puts attributes in the wrong tier, and it is the more likely one in practice: guessing that subjective attributes are around 0.3 is easier than deciding whether loudness is semi-objective or subjective. We therefore mis-tier a fraction ϕ of attributes, drawn at random and reassigned to a different tier, and sweep ϕ from 0 to 1 with the truth held fixed, so the only thing degrading is the annotation (Table XIX).

This replaces a misspecification surface that could not have shown anything. Sweeping a *homogeneous* assumed crossover against a homogeneous true one, as one might naturally do, cannot affect selection at all: by Proposition 5 a constant $\bar{\epsilon}$ adds the same capacity term to every question and leaves the ranking exactly unchanged, so such a grid measures only the belief update. The axis that bears on selection is whether attributes are in the right tier.

The advantage decays roughly linearly and changes sign between $\phi = 0.6$ and $\phi = 0.8$; linear interpolation puts the

break-even at $\phi = 0.72$. Taken at face value that says the curator may mis-tier about fifteen of the twenty-one attributes before the correction stops paying, which is a remarkable amount of slack. We would not defend the second digit. At $n = 300$ and accuracies in the low single digits each cell rests on a handful of correct identifications, the spread across independent corruptions at a fixed ϕ is of the same order as the effect being measured—at $\phi = 0.5$ the three corruptions give +1.00, +4.33 and +2.33, which is why that row sits above its neighbours—and the honest reading is that the break-even lies somewhere between $\phi = 0.6$ and $\phi = 0.8$ rather than at any particular point in between.

The qualitative conclusion survives that caveat, and it is the one that matters for deployment: the annotation is not a precision instrument. Getting a *majority* of attributes into the right tier is sufficient, and the method degrades gracefully rather than falling off a cliff. Even at $\phi = 1$, where every attribute has been moved, RAIG loses by only 0.56 pp—because reassignment is to one of the two other tiers rather than to an adversarial one, so some of the objective/subjective contrast survives by chance. The adversarial column of Table X, where the assignment is inverted rather than scrambled, is the genuinely damaging case, and it costs 13.45 pp.

F. Other Domains: the Bias Is Absent, the Benefit Is Not

The bias characterised in Section VII-A does not replicate in the two non-music catalogues, and we report that plainly because it is a confirmation of the mechanism rather than a failure of the result.

On mushroom, the Kruskal–Wallis test across tiers gives $H = 1.75$ ($p = 0.416$) and Kendall’s τ_b is +0.006 ($p = 0.974$): no relationship whatsoever, with tier means of 0.780, 0.976 and 0.803 bits. On dermatology the omnibus test is marginal ($H = 6.21$, $p = 0.045$) but the rank correlation is *positive*, $\tau_b = +0.135$ ($p = 0.335$), and the tier means are non-monotone—0.797, 0.969, 0.762 bits for objective, semi-objective and subjective—so the effect there is that the middle tier splits best, which is not the pattern the bias predicts and does not favour our method. Neither catalogue contains a continuous attribute to discretise, and the mechanism we blame is discretisation acting on perceptual quantities. Predicting the absence of an effect and then observing it is stronger evidence for the mechanism than a third positive replication would have been.

What happens next is the interesting part, and it was not predicted (Table XX).

On mushroom, reliability-aware selection changes nothing: 10.3% against 10.5% at $\lambda = 1.0$, McNemar $p = 1.00$. On dermatology it more than doubles accuracy, 73.0% against 34.7%, and the effect grows as users get noisier. Two catalogues with no measurable bias, and diametrically opposite outcomes. The correlation between informativeness and reliability evidently is not what decides whether the correction pays, and the next subsection identifies what does.

G. When Reliability-Aware Selection Pays

Two properties of a catalogue matter, and they are separate.

TABLE XX

CROSS-DOMAIN COMPARISON, TOP-1 ACCURACY (%) AT $T = 20$, $n = 600$ PAIRED TRIALS PER CELL. THE BIAS IS PRESENT ONLY IN MUSIC, YET RELIABILITY-AWARE SELECTION IS INERT ON MUSHROOM AND TRANSFORMATIVE ON DERMATOLOGY. THE LAST ROW GIVES THE EXACT MCNEMAR p -VALUE FOR RAIG-TIERED AGAINST IG+SOFT-UNIFORM IN THAT COLUMN.

Method	Mushroom			Dermatology		
	$\lambda=0.5$	1.0	1.5	$\lambda=0.5$	1.0	1.5
Random+soft	0.3	0.2	0.0	26.2	9.7	3.0
IG+hard	27.7	8.0	1.0	47.0	20.0	7.8
IG+soft-uniform	31.7	10.5	2.3	84.5	34.7	6.0
IG+soft-objective	1.7	1.2	0.8	18.5	17.3	15.8
RAIG-tiered	31.5	10.3	2.7	92.7	73.0	48.3
RAIG-oracle	35.7	10.3	3.8	94.5	73.0	46.3
McNemar p	1.00	1.00	0.85	$6 \cdot 10^{-7}$	$3 \cdot 10^{-52}$	$5 \cdot 10^{-71}$

TABLE XXI

THE RELIABLE-SUBSET AUDIT. FOR EACH CATALOGUE WE REPORT THE TOP-1 ACCURACY CEILING ATTAINABLE USING ONLY THE ATTRIBUTES A USER CAN ANSWER (OBJECTIVE AND SEMI-OBJECTIVE TIERS), COMPUTED BY COUNTING DISTINCT ATTRIBUTE VECTORS. THIS IS A PROPERTY OF THE CATALOGUE, COMPUTABLE IN SECONDS AND REQUIRING NO USERS, AND IT PREDICTS THE CROSS-DOMAIN OUTCOME THAT THE BIAS MEASUREMENT DOES NOT.

Catalogue	all attributes		reliable only	
	floor	ceiling	floor	ceiling
Music	16.23	93.7%	15.35	37.1%
Mushroom	12.99	100.0%	8.55	0.3%
Dermatology	8.52	100.0%	8.24	70.8%

Correlation determines whether classical information gain is actively harmed. In music, the selector seeks out unreliable questions: it asks at mean true crossover 0.225 against a catalogue average of 0.098 (Table XII). In dermatology it does not: it asks at 0.236 against a catalogue average of 0.240, which is to say it is behaving no worse than chance with respect to reliability. In mushroom it is mildly *better* than chance, asking at 0.170 against an average of 0.197. This is exactly the ordering the bias measurement predicts, and it confirms that the bias measurement is measuring what we say it does.

Heterogeneity, plus the availability of reliable information, determines whether reliability-aware selection can gain. Even when classical information gain is not being actively misled, a selector that knows which attributes are answerable can steer towards them—provided those attributes still carry enough information to finish the job. That proviso is measurable from the catalogue alone, before any user is involved and before any session is run, and it is what separates dermatology from mushroom (Table XXI).

In dermatology, the nine objective and semi-objective attributes alone place 70.8% of cases in a singleton equivalence class. Steering towards them costs almost nothing, and RAIG-tiered duly reaches 73.0%—a figure that sits essentially *at* the reliable-subset ceiling, slightly above it because the method still spends 22% of its turns on subjective attributes rather than banning them. In mushroom, the fifteen reliable attributes place 0.3% of specimens in a singleton class: the information needed to identify a mushroom is in its odour and its colours,

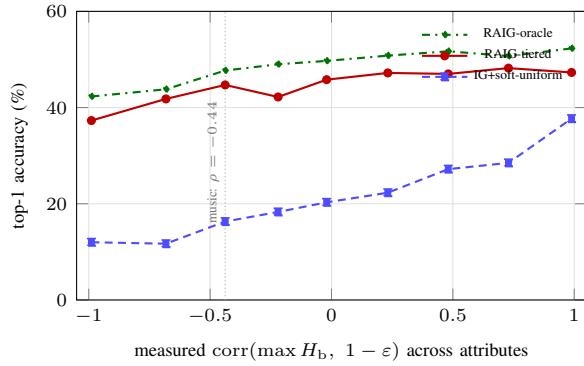


Fig. 4. The advantage as a continuous function of the informativeness/reliability correlation, on a synthetic catalogue family with everything else held fixed. Classical information gain degrades monotonically as the correlation turns adversarial, from 37.7% at $\rho = +0.99$ to 12.0% at $\rho = -0.99$. Reliability-aware selection is close to flat over the same range (47.3% to 37.3%). The correlation therefore measures how badly classical selection is *harm*ed, not whether the correction is *available*; the dotted line marks the correlation measured on the music catalogue.

and a selector that avoids them cannot succeed. RAIG-tiered accordingly shifts its question mix only slightly (47% objective against IG’s 35%) and gains nothing. Music sits between, at 37.1%.

This gives a deployment test that costs a database query. *Compute the identifiability ceiling of the reliable attribute subset. If it is high, reliability-aware selection will help substantially; if it is near zero, the unreliable attributes are load-bearing and no selection rule can avoid them—though the belief update should still model their noise.* It also explains the restricted-attribute baseline’s behaviour without appeal to tuning: on music the objective-only ceiling is 0.7%, and IG+soft-objective scores 0.33%.

The correlation as a continuous axis. Three catalogues give three points on the correlation axis, which is not enough to see its shape. A synthetic family makes the axis itself the independent variable: attributes are drawn with a controlled range of split balance and assigned crossover probabilities on a fixed grid from 0.02 to 0.35 at a target rank correlation with that balance. Everything else—catalogue size, attribute count, budget, the three-tier bucketing of the drawn crossovers—is held constant, so the only thing varying across the sweep is the alignment between informativeness and reliability. Fig. 4 reports the result over nine settings, $n = 600$ paired trials each.

The sweep contains no break-even point. RAIG-tiered leads at every setting, from +9.67 pp at $\rho = +0.99$ —where the informative attributes are also the reliable ones and there is, on the correlation account, nothing to correct—to +30.17 pp at $\rho = -0.68$. That is the same lesson the cross-domain results gave, now on a controlled axis: what the correlation governs is the *denominator*, how far classical information gain falls, not whether reliability-aware selection has anything to offer. And the reason the correction is always available here is the one the previous paragraph identifies—in this family twelve of the twenty attributes fall in the reliable tiers and place 76% to 94% of items in a singleton class, so steering towards them

TABLE XXII

THE ABSTENTION CHANNEL. MILD IS $\delta = (0.02, 0.10, 0.20)$ BY TIER, STRONG IS $(0.05, 0.20, 0.40)$; THE ERROR CHANNEL IS HETEROGENEOUS $\lambda = 1.0$ THROUGHOUT, $n = 450$ PAIRED TRIALS. “ANSWERED” IS THE FRACTION OF CONSUMED TURNS THAT RETURNED AN ANSWER RATHER THAN AN ERASURE, AND IS A CONSEQUENCE OF WHICH QUESTIONS THE SELECTOR CHOSE.

Abstention	Method	acc. (%)	answered	subj. %
none	IG+soft-uniform	2.44	1.000	53.4
	RAIG-tiered	7.78	1.000	2.8
	RAIG+erasure	7.78	1.000	2.8
mild	IG+soft-uniform	1.56	0.849	54.3
	RAIG-tiered	5.33	0.930	1.8
	RAIG+erasure	4.00	0.932	1.2
strong	IG+soft-uniform	1.11	0.692	55.6
	RAIG-tiered	4.00	0.854	1.4
	RAIG+erasure	3.56	0.862	0.5

is affordable at every ρ .

The practical reading combines the two experiments. A catalogue with an adversarial correlation *and* an answerable reliable subset is the best case, and it is where the music catalogue sits. A catalogue with an answerable reliable subset but no correlation still benefits, and that is dermatology. A catalogue whose reliable subset cannot identify anything does not benefit whatever its correlation, and that is mushroom.

H. Abstention

Section III-G derives the acquisition function for a user who may decline to answer, (10), and attaches a prediction to it: if abstention correlates with subjectivity in the same direction as error, the two penalties compound and the bias should be *larger* than the pure-BSC analysis suggests. Predictions of that kind belong in an experiment rather than in a future-work list, so we implement the channel and test it. Each attribute additionally returns an erasure with probability δ_f , tied to its tier on the reasoning that a user who cannot judge an attribute reliably is also the one most likely to decline it; an erased turn is consumed but produces no belief update. Three selectors are compared: classical information gain, blind to both; RAIG, which models error but not abstention; and RAIG+erasure, which models both.

Table XXII reports the outcome. There are two results, and the first is a refutation of our own prediction.

The absolute advantage shrinks rather than grows. RAIG-tiered leads IG+soft-uniform by +5.33 pp with no abstention, +3.78 pp under mild abstention and +2.89 pp under strong abstention. Measured on the paper’s headline scale, the penalties do not compound; they compress, because abstention pushes every method towards the floor and there is progressively less accuracy to win. The prediction of Section III-G is wrong as stated, and we let it stand in the theory section rather than quietly deleting it, because the version that survives is worth having: on a *relative* scale the compounding is real and monotone, with RAIG’s accuracy $3.18\times$ IG’s without abstention, $3.43\times$ under mild and $3.60\times$ under strong. The right conclusion is that abstention makes classical selection

relatively worse and absolutely no more distinguishable, which is a weaker claim than the one we made in advance.

Modelling abstention explicitly buys nothing here. RAIG+erasure is indistinguishable from RAIG-tiered with no abstention (identical, as it must be when $\delta \equiv 0$) and slightly *worse* under both mild (-1.33 pp, $p = 0.24$) and strong (-0.44 pp, $p = 0.80$) abstention. Neither difference is significant, so the finding is an absence rather than a reversal, but the point estimate is in the wrong direction and the mechanism is instructive. Because δ_f is tied to the tier in the same order as ε_f , the error-based discount has already steered the selector away from the high-abstention questions: RAIG-tiered gets 93.0% of its turns answered under mild abstention against classical selection’s 84.9%, without modelling abstention at all. The extra $(1 - \delta)$ factor then applies a second discount to attributes that are already being avoided, driving the subjective share of questions to 0.5% at strong abstention and narrowing the question pool towards the reliable-subset ceiling of Section VIII-G—the same over-correction that produces the clean-condition collapse.

The practical implication is that a deployed system facing abstention should model it in the *belief update*, where an erasure must not be treated as evidence, and can safely leave the selector to the error term alone—provided abstention and error are ordered the same way across attributes. Where they are not, (10) is the correct criterion and this experiment says nothing against it.

I. Correlated Answer Errors

The channel of Section III draws every answer independently at rate ε . A real listener who misjudges “is it energetic?” is likely to misjudge it the same way when asked again, so independence is the modelling assumption most likely to be flattering to us, and it should be attacked directly rather than listed as a limitation.

We attack it with a deliberately harsh alternative. The first time a *subjective* attribute is answered wrongly in a session, that attribute is marked corrupted for the remainder of the session and every later question about it is forced wrong; objective and semi-objective attributes keep independent draws, since simple slips rather than conceptual confusion are the plausible failure there. This turns a rate- ε channel into a per-session absorbing one on exactly the attributes classical selection prefers. Both variants run inside the same loop and consume the same random stream in the same order, so the independent and correlated columns are paired trial for trial and any difference between them is the correlation and nothing else.

The result, in Table XXIII, strengthens rather than weakens the paper’s conclusion.

Correlating the errors costs IG+soft-uniform 0.78 pp—seven trials that it won under independent errors and loses under correlated ones, and not a single trial in the other direction—while RAIG-tiered’s outcome is *bitwise identical* across the two variants: zero discordant pairs out of 900. The paired advantage therefore grows from +5.00 pp to +5.78 pp, and its p -value improves by two orders of magnitude.

TABLE XXIII
INDEPENDENT AGAINST CORRELATED ANSWER ERRORS,
HETEROGENEOUS $\lambda = 1.0$, $n = 900$, PAIRED TRIAL FOR TRIAL ACROSS
ALL FOUR CELLS. CORRELATING ERRORS ON SUBJECTIVE ATTRIBUTES
COSTS CLASSICAL SELECTION 0.78 PP AND COSTS RELIABILITY-AWARE
SELECTION EXACTLY NOTHING.

Method	independent	correlated	difference
IG+soft-uniform	2.78	2.00	-0.78 pp ($p = 0.016$)
RAIG-tiered	7.78	7.78	$+0.00$ pp ($p = 1.00$)
Δ (RAIG – IG)	+5.00 pp	+5.78 pp	
McNemar p	3.6×10^{-7}	3.1×10^{-9}	

The asymmetry is not luck, and it follows from Table XII. The absorbing mechanism can only fire on subjective attributes. Classical selection spends 53.9% of its budget there and is exposed to it on most turns; RAIG-tiered spends 2.9% and is almost never exposed, and in these 900 sessions it was never exposed twice to the same subjective attribute, which is what a repeat requires. A selector that avoids the attributes on which users are unreliable is, as a side effect, also robust to their errors being persistent rather than fresh—which is a property worth having, since persistence is the more realistic model and the harder one to estimate.

We do not claim this exhausts the question. Errors could correlate *across* attributes rather than within one—a listener with an unusual notion of “energetic” probably also has an unusual notion of “danceable”—and that structure would penalise any method that clusters its questions within a tier, including ours. Testing it would require a user-level latent-trait model that we do not have data to fit.

J. An Online Alternative to the Tiers, and Its Catalogue-Size Dependence

Section VIII-B removes the three assumed tier values by learning a crossover vector *offline*, from six hundred already-completed sessions, before the vector is ever used. A natural sharper version of the same idea is to skip the offline pass entirely: re-estimate the per-attribute crossover *online*, from the trials the deployed system is already running, with no tier annotation assumed even as a cold-start value. We built and tested exactly this variant, **RAIG-streaming**, and report both what happened and why, because the failure is as informative as a success would have been.

The mechanism. Every attribute starts at a single neutral crossover, $\hat{\varepsilon} = 0.10$, no tiers involved. Between batches of paired trials—not inside the per-turn loop itself, which would couple the estimate to a single session’s own noise—the same attribute-pooled expected-disagreement statistic Section VIII-B computes offline is folded into a running per-attribute total, and the estimate used by the NEXT batch is the resulting mean, shrunk toward the neutral prior with a pseudo-count of sixty so that an attribute with little evidence cannot swing on a handful of noisy readings. A fraction of turns, 20%, is forced to a uniformly random available question regardless of the current estimate, mirroring the forced-exploration fix Section VIII-B’s offline estimator already needed for the identical reason.

TABLE XXIV

STATIC TIERS AGAINST THE ONLINE ALTERNATIVE, PRIMARY CONDITION, $T = 70$, $n = 900$. Δ AND p ARE AGAINST RAIG-TIERED, EXACT MCNEMAR, HOLM-CORRECTED WITHIN THIS FAMILY.

Method	top-1 (%)	Δ (pp)	p (Holm)
IG+soft-uniform	27.11	-11.44	5.1×10^{-9}
RAIG-streaming	18.56	-20.00	5.0×10^{-27}
RAIG-streaming+stop@0.5	19.78	-18.78	5.2×10^{-24}
RAIG-streaming+stop@0.7	18.00	-20.56	1.2×10^{-28}
RAIG-streaming+stop@0.9	17.78	-20.78	1.2×10^{-28}
RAIG-tiered	38.56	—	—

On the full catalogue, it loses, to a diagnosed mechanism rather than to noise. At the primary condition and $T = 70$ (Table XXIV), RAIG-streaming reaches 18.56% against RAIG-tiered’s 38.56%—a 20.00 pp deficit, exact McNemar $p = 5.0 \times 10^{-27}$ —and even trails plain classical information gain’s 27.11% by 8.56 pp ($p = 4.4 \times 10^{-6}$). The cause is a feedback loop the fixed tiers cannot experience because they never move. *genre* alone compiles to 113 of the catalogue’s 193 questions and is genuinely the most reliable attribute in it; the moment the running estimate correctly notices this, the selector asks it more, which supplies more evidence for exactly that attribute and none for the other nineteen, whose own estimates stay noisy and drift toward the unreliable end of the clip range from thin, badly-timed samples. Measured directly over these 900 sessions, *genre* takes 58.6% of every question RAIG-streaming asks, against 36.0% for the curator’s fixed tiers and 24.6% for classical information gain at the same operating point—a share no fixed annotation ever produces, because a fixed annotation cannot compound the way a live one can. An earlier, unregularised attempt (no shrinkage, 10% exploration) put *genre*’s share at 78.7% and reached only 5.9% accuracy; the shrinkage-and-exploration fix reported here roughly triples that, but does not close the gap to the tiers at this scale.

Raising the cap alone, on the tiers, is the clean win in this experiment. RAIG-tiered itself improves from 26.56% at the budget of Table IX to 38.56% simply by raising T from 40 to 70 with nothing else changed—an operating-point change, not a methodological one, and the only unambiguous gain reported in this subsection.

Early stopping at $T = 70$ behaves exactly as Section VII-C predicts it will at $T = 20$: it mostly does not fire. A confidence threshold of 0.90 crosses in 1.1% of RAIG-streaming sessions by turn seventy; 0.70 in 2.0%; 0.50 in 3.6%. None of the three changes accuracy by a significant amount relative to running the streaming estimator to completion (Table XXIV, all $p_{\text{Holm}} = 1.00$). Forty additional questions past the entropy floor still mostly does not buy the confidence a stopping rule needs to act on; it buys accuracy at a fixed budget instead, which is the mechanism Table XXIV already reports.

The deficit is not fixed: it is catalogue-size dependent. Section VII-F already reports that the RAIG-over-IG advantage survives at every catalogue size tested. We repeated the online-versus-static comparison specifically, on an independently drawn 5,000-item subset (identifiability ceiling 99.5%, floor

TABLE XXV

THE ONLINE ESTIMATOR AGAINST STATIC TIERS ON A 5,000-ITEM SUBSET, ONE INDEPENDENT DRAW, $n = 900$, PRIMARY CONDITION. p IS THE EXACT MCNEMAR TEST BETWEEN RAIG-STREAMING AND RAIG-TIERED AT THAT COLUMN.

Method	$T=40$	$T=60$	$T=80$	$T=100$	$T=200$
IG+soft-uniform	41.33	56.67	58.44	58.44	58.00
RAIG-tiered	61.33	69.56	71.44	72.89	72.67
RAIG-streaming	47.89	64.89	71.00	72.11	73.89
p (streaming vs. tiered)	—	—	0.85	0.71	0.52

12.28 bits) with the budget extended to $T = 200$ (Table XXV). The pattern reverses: by $T = 80$ the two are statistically indistinguishable (71.00% against 71.44%, $p = 0.85$), and the gap stays indistinguishable from zero through $T = 200$ (73.89% against 72.67%, $p = 0.52$)—RAIG-streaming’s point estimate is even fractionally higher at the longest budget tested, but not significantly so. We do not claim the online estimator wins at small scale; we report that the deficit it shows at full scale is not a fixed property of the method, and that a smaller identification task gives the same online estimator enough of its own budget to reach parity with an annotation it was never given.

Reading the two results together. What the online estimator lacks relative to the offline one of Section VIII-B is repetition: the offline estimator replays the same fixed logged sequence eight times, each pass recomputing belief under a better crossover guess; the online one sees every trial once, under whichever guess happened to be current when that trial ran. On a catalogue small enough, or a budget long enough, that a single pass supplies adequate coverage of every attribute, this costs little. On the full catalogue at $T = 70$ it costs a great deal, because *genre*’s dominance of the question pool gives the feedback loop room to run before the other nineteen attributes accumulate enough evidence to compete. We report this as a diagnosed limitation with a concrete next step—an online estimator that replays its own most recent batch a small, fixed number of times before committing to the next one, borrowing the offline estimator’s repetition without its six-hundred-session latency—rather than as a closed question.

IX. DISCUSSION

A. What This Paper Establishes

Four things, the first three in decreasing order of confidence and the fourth the one that answers the objection this work is most exposed to.

That classical information gain systematically solicits the questions users answer worst, in catalogues of a particular shape, is established by direct measurement of selector behaviour and does not depend on any assumed ε . On the music catalogue, a classical selector asks questions whose annotated crossover averages 0.225 against a catalogue average of 0.098, filling 53.9% of a twenty-question budget with the tier a listener judges least reliably, while a uniformly random selector faces 0.097, which is the catalogue average. Whatever one believes about the numeric values of ε , the ordering of the annotation is what drives that contrast, and the contrast is large.

That *the effect is manufactured by discretisation rather than inherited from the domain* is established by three independent lines that agree: Proposition 4, which shows quantile binning fixes apparent informativeness at $H_b(1/k)$ irrespective of semantics; the measurement that binned attributes carry mean maximum information gain 0.861 against 0.618 for natively categorical ones; and the observation that refining the binning to four equal-population bins *strengthens* the measured bias, from $\rho = -0.44$ to $\rho = -0.61$. The absence of the effect in two catalogues with no continuous attributes to bin is the prediction this account makes, and it holds.

That *correcting the acquisition function helps, and does so from an annotation cheap enough to be deployable*, is established at $n = 900$ paired trials with exact tests and multiplicity control, and—more importantly—is shown to survive the assumption it rests on: over 60 worlds drawn from priors on the tier values, the fixed deployed system wins in 60 of 60, with a minimum advantage of +1.33 pp. Three coarse tiers recover roughly 72% of what perfect per-attribute knowledge would deliver, and a two-tier annotation recovers 78% of the three-tier gain (Section VII-G).

That *the three assumed constants are not load-bearing* is answered three ways, in increasing strength: the conclusion survives a distribution over those constants; it survives a majority of attributes being placed in the wrong tier; and the constants are not required at all. EM over 600 logged sessions of the already-deployed incumbent—no annotation, no labels, and the target item latent throughout—recovers a per-attribute crossover vector that matches the curator’s annotation to a dead heat, 38 discordant pairs in each direction out of 900 (Section VIII-B).

Two of the model’s structural assumptions were attacked directly, and the outcomes point in opposite directions. Relaxing independence across turns *strengthens* the result: making subjective errors persist for the rest of a session costs classical selection 0.78 pp and costs reliability-aware selection exactly nothing, because a selector that avoids unreliable attributes is not exposed to their errors being persistent (Section VIII-I). Adding abstention, by contrast, refutes a prediction we made in advance: we expected the penalties to compound and the absolute advantage to grow, and it shrinks (Section VIII-H). Only the relative advantage compounds. We report the failed prediction because a paper that only tests the predictions it expects to confirm is not testing anything.

B. What It Does Not Establish

We claim no algorithmic novelty. The acquisition function is the mutual information between the target and the observed answer under a binary symmetric channel; it is in Cover and Thomas [3, Ch. 7], and a reader who finds (6) unsurprising is reading it correctly. The contribution is that the observation model in some deployed attribute catalogues is heterogeneous in a specific and adversarial way, that the heterogeneity is partly an artefact of preprocessing, that a learned policy trained without a noise model inherits the blind spot, and that the correction needs far less knowledge about users than one would expect. Those are empirical claims, and they are what the experiments are for.

Nor have we measured ε on people, and it is worth being precise about what the log-based estimator of Section VIII-B does and does not settle. It establishes identifiability and sufficiency: a per-attribute crossover vector can be recovered from six hundred sessions of ordinary use without annotation, labels or ground truth, and recovering it is enough to match a hand-annotated one exactly. It does not establish that real users behave this way, because the logs it consumes were generated by a simulator whose users err at rates drawn from the annotation—so what EM recovers is, at bottom, the structure we put in. The same caveat applies to the ensemble over tier priors (Section VIII-A), the corruption sweep (Section VIII-E) and the repeated-item measurement (Section IV-B). Together they establish that the conclusion is not an artefact of three particular numbers. None of them tells a practitioner what ε is in their domain, and only the study of Appendix A would.

Finally, the two negative results on catalogue-derived specifications should not be softened, because together they say something the positive results do not. RAIG-empirical, which infers reliability from the rate at which repeated items disagree with each other, is beaten by the three-tier annotation by 6.33 pp and is no better than classical information gain. RAIG-psycho, which computes the crossover of each binned attribute from the position of its own cut points, fails at every setting of its two parameters (Section VIII-C). Both measure a property of the *catalogue*, and both are defeated by the same thing: a listener cannot name the key of a song, not because the boundary between keys is fuzzy or because the catalogue disagrees with itself about it, but because they never learned to name keys. Answerability is a fact about people. What succeeds—the curator’s tiers, and the vector learned from logs—succeeds because each is, in the end, an observation of people rather than of data.

C. A Deployment Test

The cross-domain results yield a procedure a practitioner can run before committing to anything, and it is the most transferable output of this work.

- 1) *Is there anything to correct?* Compute per-attribute maximum one-versus-rest entropy and compare it across a coarse answerability annotation. A Kruskal–Wallis test on three tiers is the right instrument; a Spearman correlation on three tied levels is not. If informativeness concentrates on the hard-to-answer tier, classical selection is being actively misled.
- 2) *Is the correction affordable in information?* Compute the identifiability ceiling of the reliable attribute subset by counting distinct attribute vectors. This costs one database query. If the ceiling is high (dermatology, 70.8%), reliability-aware selection will help substantially even when step 1 finds no bias. If it is near zero (mushroom, 0.3%), the unreliable attributes are load-bearing and no selection rule can route around them.
- 3) *Is the channel actually noisy?* Reliability weighting is a correction for a noisy channel and is harmful when the channel is clean: in our clean condition RAIG-tiered loses to classical selection by more than fifty points at

$T = 20$, because it breaks ties towards attributes that cannot by themselves separate the catalogue. Switch it off, not down, when answers are reliable.

- 4) *Annotate coarsely and ship, then let the system re-estimate.* Two tiers recover 78% of what three tiers deliver, and three tiers recover about 72% of what per-attribute truth would deliver. Once the system is running, that residual is addressable without a measurement campaign: EM over its own logs recovers a per-attribute vector that matches the annotation (Section VIII-B). The order of operations is therefore curator first, logs second, user study only if the first two disagree.

Steps 1 and 2 are computable from the catalogue alone. Step 3 requires knowing roughly how noisy users are, which is the one thing that genuinely needs observation—and Section VIII-D locates the threshold.

D. Threats to Validity

Simulated users. Every session in this paper is played by a simulator drawing from a binary symmetric channel. This is the central threat and we do not minimise it. Its severity is bounded in three ways: the deployed system never sees the true ε ; the conclusion is reported across a distribution of true ε rather than at a point; and two structural assumptions of the channel—that every turn is an independent draw, and that a question is always answered—are attacked directly in Sections VIII-I and VIII-H. Three things remain unaddressed. Whether real answer errors are well described by *any* per-attribute rate is the deepest of them, and only the study of Appendix A can settle it. The channel is assumed *symmetric*, whereas real users may well have a bias—readier to say “yes” than “no” when unsure—which an asymmetric binary channel would capture and which our formulation does not. And errors are correlated within an attribute in Section VIII-I but never *across* attributes, though a listener with an unusual notion of “energetic” plausibly has an unusual notion of “danceable” too; that structure would penalise any method that concentrates its questions within a tier, ours included.

One primary catalogue. The headline comparison is on music. We add two further catalogues and a synthetic family precisely because one catalogue cannot support a general claim, and the resulting scope condition is stated as a condition rather than as a universal.

Greedy selection. We select greedily throughout. Greedy optimality under noise is established for a single common crossover probability [7] and, as Remark 1 notes, does not obviously extend to heterogeneous per-attribute capacities. Holding the policy class fixed at the one deployed systems use is what isolates the acquisition function; it is not a claim that greedy is optimal here.

Static reliability. $\hat{\varepsilon}$ is fixed for the session. A real system could estimate a user’s per-attribute reliability online from answer consistency, which would make the specification problem largely disappear—and would introduce an exploration/exploitation trade-off this paper does not address.

Simulator–method coupling. The simulator and the method share the functional form of the noise model. This is standard

in the noisy-search literature but it is a real limitation: the ensemble experiment breaks the parameter coupling but not the functional-form coupling, and the abstention experiment is the only place where the true channel has a component the baseline selector cannot represent.

E. Future Work

The obvious next step is the user study, and it is specified rather than gestured at. Beyond it, three directions follow from what the experiments showed rather than from what would be conventional to list.

Per-user reliability estimation is the most valuable, and is the natural extension of what Section VIII-B already does. That estimator pools every session into one crossover vector, which is the right object when the population is homogeneous and the wrong one when it is not: a musician and a casual listener do not share a crossover for “what key is it in”. Estimating $\hat{\varepsilon}_f$ per user, from within-session answer consistency and a population prior, is a hierarchical version of the same EM and would attack the variation this paper’s model cannot express. It also introduces an exploration/exploitation trade-off—questions asked to learn about the user rather than about the item—that Section VIII-B shows is not free, and Section VIII-J shows can be worse than not free: we tried the simpler, non-hierarchical version of this idea—one crossover vector, updated *online* from the deployment’s own trials instead of offline from a fixed log—and it loses badly on the full catalogue to a feedback loop the offline estimator’s repeated replay does not allow (genre monopolises 58.6% of every question asked once the running estimate correctly favours it, starving the other nineteen attributes of the evidence needed to estimate them), though the deficit disappears on a smaller catalogue with a longer budget. A per-user version would face the identical loop at the scale of a single session rather than the whole deployment, which is a reason for pessimism about a naive implementation and a concrete requirement—replay, or an explicit optimism bonus for under-sampled attributes, not exploration alone—for a serious one.

Non-myopic selection under heterogeneous capacities is the most interesting theoretically. Corollary 1 makes the difficulty concrete: with per-attribute capacities, a horizon-optimal policy may wish to reserve a high-capacity attribute for a later turn when the belief can exploit it, and a greedy rule cannot express that preference.

Finally, the discretisation result suggests attacking the problem one stage earlier. If quantile binning is what manufactures the bias, then a reliability-aware *discretiser*—one that chooses bin boundaries to maximise mutual information through the channel rather than to equalise populations—would prevent it rather than compensate for it. Nothing in this paper requires the binning to be given.

X. CONCLUSION

Expected information gain is the default acquisition function for interactive identification, and it was derived for an oracle that answers exactly. When the oracle is a person, the criterion has a blind spot with a specific shape: it scores a question by

how evenly it divides the belief, and vagueness is what makes a split even. In a catalogue that mixes quantile-binned perceptual descriptors with skewed categorical metadata, the criterion therefore seeks out the questions users answer worst—not as a failure of tuning but as a consequence of what it optimises, sharpened by a preprocessing step that fixes apparent informativeness at a value determined by the bin count rather than by the attribute.

The correction is a single line, and it is textbook: score the question by the mutual information between the target and the observed answer through a per-attribute binary symmetric channel. What we have tried to establish is not that this quantity is new but that it is needed, and that the knowledge it requires is cheap enough to be had. Two coarse tiers recover most of the benefit; the annotation may be wrong for a majority of attributes before it stops paying; across sixty worlds drawn from priors on the tier values a system shipping the fixed point estimates wins in sixty; and a two-line audit of the catalogue predicts in advance whether any of this will help. The three numbers need not even be assumed: EM over six hundred logged sessions of the already-deployed incumbent, with the target item latent and nothing annotated, recovers a crossover vector that matches the curator’s exactly.

We have also been explicit about the boundaries. Reliability weighting is a correction for a noisy channel, and on a clean one it is not merely useless but harmful. It requires a reliability model, and inverting that model inverts the result. The two routes that would have supplied one from the catalogue alone—repeated-item instability, and a psychophysical model of the bin boundaries—both fail, and fail for the same reason: they measure the data, and answerability is a property of the people. The route that does work without a curator is to watch those people, which is what the log-based estimator does and why it succeeds where the catalogue-derived ones do not. And the parameter has still not been measured on real people, only on simulated ones. That is the piece of work this paper specifies and does not do, and it is the piece that would close the argument.

APPENDIX A

A PROTOCOL FOR MEASURING ε_f FROM USERS

The most exposed assumption of this paper is the tier annotation (0.03, 0.15, 0.30), and four experiments reduce the exposure without removing it. Section IV-B measures an adjacent quantity from the catalogue; Section VIII-A shows the conclusion survives a distribution over the values; Section VIII-E shows how wrong the tier assignment can be before the correction stops paying; and Section VIII-B dispenses with the values entirely, recovering them from interaction logs.

None of that is the same as measuring ε_f on people, and the last of the four is the one whose limits are easiest to overstate. Its logs come from a simulator whose users err at rates drawn from the annotation, so it establishes that the parameter is *identifiable* from logs of that size and that recovering it *suffices*, not that human error rates have this shape. Only observation of people can establish that. We did not run a human study, and rather than list that as a limitation

we specify the study, so that the gap is a piece of work someone can do rather than a hole in the argument.

Design

The estimand is not “how often is a user wrong”, which is unidentifiable without ground truth about what the user is thinking. It is the crossover probability of the channel between the item a user has in mind and the answer they give: $\varepsilon_f = \Pr[\tilde{a}_f \neq a_f(c) \mid c]$, where $a_f(c)$ is the catalogue’s value and $\tilde{a}_f$ the user’s report. This is identifiable if the item is known to the experimenter and unknown to be under test by the participant.

Task. A participant is shown a track—title, artist, and a thirty-second audio excerpt—and asked the same one-versus-rest binary questions the deployed system would ask, in randomised order. Because the catalogue value $a_f(c)$ is known, each answer is directly scoreable, and $\hat{\varepsilon}_f$ is the per-attribute error rate with a Wilson interval [33].

Recall versus perception. A user of the deployed system answers from memory, not from an excerpt. The excerpt condition therefore measures a lower bound on ε_f . We would run both arms: a *perception* arm with the excerpt available, and a *recall* arm in which participants nominate tracks they claim to know well and answer without hearing them. The contrast between the arms is itself the interesting quantity for objective attributes, where the excerpt makes the answer nearly free but recall does not.

Sample size. To separate the annotated tiers we need to distinguish 0.03 from 0.15 and 0.15 from 0.30. At $\varepsilon = 0.15$, a Wilson interval of half-width 0.05 requires roughly 200 answers per attribute. With 20 attributes and 30 questions per participant, that is 130–140 participants, or fewer if each answers about several tracks. A pilot at $n = 25$ would not resolve individual attributes but would resolve the *tier contrast*, which is what the method actually consumes: pooling within tiers gives roughly 1,300 answers per tier at $n = 25$, ample to reject tier equality.

Analysis. The per-attribute rate is a proportion with a Wilson interval; the tier contrast is a mixed-effects logistic regression with a random intercept per participant and per track, since answers from the same person and about the same track are not independent. The pre-registered test is the tier main effect. The secondary analysis regresses $\hat{\varepsilon}_f$ on (i) the annotated tier, (ii) the measured catalogue discordance d_f of (11), and (iii) the bin count of the attribute, which is the direct test of the psychophysical mechanism of Section IV-E.

What the Study Would and Would Not Settle

It would settle whether the three-tier ordering is real and roughly how far apart the tiers are, which is the only property of the annotation that the method uses: by Proposition 5 only the heterogeneity does any work, by Section VIII-B an estimate that is correctly ordered but badly calibrated performs as well as the truth, and by Section VIII-E the tier assignment may be wrong for most attributes before the correction stops paying. The study needs to establish an ordering, not three decimal places, and it is powered accordingly.

It would not settle the point values for a different catalogue, a different population, or a different interface, and it should not be read as doing so. The transportable claim in this paper is conditional and has two parts, which Section VIII-G separates. Where answerability is heterogeneous *and* negatively aligned with apparent informativeness, classical selection is actively misled and the correction pays most. Where it is heterogeneous but unaligned, classical selection is merely blind and the correction still pays, provided the answerable attributes carry enough information to finish the job. Where they do not, no selection rule can route around them. Three coarse tiers capture most of the available gain in the first two cases, and nothing helps in the third.

APPENDIX B REPRODUCTION

Every number in this paper is produced by a script in the released repository and stored as JSON, and the tables are checked against those files mechanically rather than transcribed by hand. The determinism requirements of Section V—single-threaded BLAS, externally supplied random streams—make the runs bitwise reproducible on a given machine and make each method’s results invariant to which other methods are run alongside it.

Two caveats are worth recording. First, results generated before the single-thread pin was introduced do not reproduce bitwise; they agree within sampling error, and where such a run is quoted we say so. Second, the budget sweep, the ε -ensemble and the cross-domain comparisons are independent runs at their own sample sizes, so their overlapping cells agree within sampling error rather than exactly. We report the overlaps rather than suppressing them.

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